

Blood pressure variability compromises vascular function in middle-aged mice

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eLife Assessment

This is an **important** study that demonstrates that blood pressure variability impairs myogenic tone and diminishes baroreceptor reflex. The study also provides evidence that blood pressure variability blunts functional hyperemia and contributes to cognitive decline. The evidence is **compelling** whereby the authors use appropriate and validated methodology in line with or more rigorous than the current state-of-the-art.

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Abstract

Blood pressure variability (BPV) has emerged as a significant risk factor for cognitive decline and dementia, independent of alterations in average blood pressure (BP). However, the impact of large BP fluctuations on neurovascular function remains poorly understood. In this study, we developed a novel murine model of BPV in middle-aged mice using intermittent Angiotensin II infusions. Radio telemetry confirmed that 24-hr BP averages in BPV mice remained comparable to controls, demonstrating BPV in the absence of hypertension. Chronic (20-25 days) BPV resulted in a blunted bradycardic response and cognitive deficits. Two-photon imaging revealed heightened pressure-evoked constrictions (myogenic response) in parenchymal arterioles of BPV mice. While sensory stimulus-evoked dilations (neurovascular coupling) were amplified at higher BP levels in control mice, this pressure-dependent effect was abolished in BPV mice. Our findings indicate that chronic BP fluctuations impair vascular function within the neurovascular complex and contribute to cognitive decline, emphasizing BPV as a critical factor in brain health.

Introduction

Dementia, the 7th leading global cause of mortality, presents as the predominant neurodegenerative complication in the elderly population. Notably, hypertension emerges as a primary risk factor for dementia^{1,2,3}. Blood pressure (BP) regulation involves a complex

interplay of central and peripheral mechanisms, finely attuned to acute and chronic stressors in maintaining physiological ranges. Perturbations in vascular function and structure (e.g., reactivity, vascular stiffness³), alterations in baroreflex sensitivity⁴, and enhanced sympathetic activation⁵ are recognized factors that disrupt BP balance. Blood pressure variability (BPV) (an index of BP *fluctuation or variation*), particularly in midlife, increases the risk for cognitive decline⁶, end-organ damage⁷, and cardiovascular events^{7,8}. Increased BPV also serves as an early predictor of hypertension⁹. BPV delineates fluctuations across various measurement periods from very short-term (beat-to-beat) to long-term (months or years¹⁰).

Given evidence that elevated BPV can precede hypertension¹¹, a modifiable risk factor in the setting of vascular cognitive impairment and dementia, understanding the drivers and cellular targets of BPV is essential. Unlike hypertension, BP fluctuations often elude detection in screening practices conducted in clinical settings, which rely on singular or averaged BP measurements¹². The absence of a gold standard for BPV indexing¹³ and limited insights into the magnitude and frequencies of impactful BP fluctuations in disease conditions further compound this issue. Additionally, the absence of suitable animal models has limited the understanding of how this overlooked variable may impact brain health. In this study, we sought to establish an innovative murine model of BPV and to interrogate the impact of large BP fluctuations on cardiovascular and neurovascular outcomes.

Cerebral blood flow (CBF) is tightly regulated by diverse signaling processes, including the interaction of cellular elements comprising the neurovascular complex, which include endothelial cells, vascular smooth muscle cells, pericytes, astrocytes, neurons, and microglia^{14,15}. Mechanisms governing CBF operate under both steady-state conditions (e.g., cerebral autoregulation (CA), humoral processes, chemoregulation) and in response to local neuronal activity (e.g., neurovascular coupling (NVC))¹⁶. Evidence suggests that sustained hypertension adversely affects the functional integrity of the neurovascular complex^{17–19}, compromises cerebral perfusion^{17,18} and contributes to cognitive decline¹⁸. However, the impact of large BP fluctuations on steady-state or activity-evoked CBF changes remains poorly understood.

Chronic hypertension leads to adaptive processes (e.g., vascular remodeling^{20,21}) and a rightward shift in the cerebral autoregulation (CA) curve²², heightening vulnerability to ischemia at low BP while shielding the brain from hyperperfusion at high pressure. In humans, hypertension and BPV have been associated with white matter hyperintensities^{23–25} and microbleeds^{26–28}. Intriguingly, the impact of BPV (prior to hypertension onset) on cerebrovascular function and neurovascular outputs has received little attention.

Using an innovative murine model of BPV, in the absence of overt hypertension, combined with *in vivo* two-photon imaging, we show that chronic BP fluctuations lead to microvascular dysfunction (e.g., enhanced parenchymal arteriole myogenic responses) and a blunted NVC response. In addition, mice subjected to chronic BPV showed poor cognitive performance. These findings underscore the pivotal role of dysregulated BP events in brain health and function.

Methods

Animals

All experiments were conducted in middle-aged (12–15-month-old) male C57BL6 mice (Jackson Laboratories) under protocols approved by the Institutional Animal Care and Use Committee of Augusta University (AU). Mice were housed at 20–22°C under a 12hr:12hr light-dark cycle with *ad libitum* access to food and water. Female mice were not included in the present study due to high

post-surgery mortality observed in 12-14-month-old mice following complex procedures. To minimize confounding effects of differential survival and to establish foundational data for this model, we restricted the investigation to male mice.

Craniotomy surgery for chronic window

Surgeries were conducted using the aseptic technique. Animals were injected with dexamethasone and meloxicam 2-4 hours before surgery to prevent edema and/or inflammation. General anesthesia was induced with isoflurane (2%) and following loss of reflex, hair was removed from the scalp and the mouse was transferred to a sterile field. A single injection of ketamine/dexdomitor (60 mg/kg/0.5mg/kg) was administered to maintain anesthesia in the sterile field. The scalp was scrubbed with betadine/alcohol (3 times each, alternating). A scalp incision was made (~1 cm). A small aluminum head holder (300 mg) was fixed to the skull using cyanoacrylate glue followed by dental cement and a small craniotomy opened over the cortex. The bone flap was removed. A glass coverslip (3 mm) was glued to a 4 mm coverslip and placed into the craniotomy with the inner glass touching the cortex. The edges of the coverslip were then sealed with cyanoacrylate glue followed by dental cement covering everything including the wound margins.

Telemetry and iPrecio pump implantation surgery

Following a 3-week recovery period after craniotomy surgery, mice were implanted with a programmable pump (iPrecio, SMP-310R) and a biotelemetry transmitter device (PA-C10, Data Science International). Mice underwent brief anesthesia in a small chamber using 5% isoflurane for induction, followed by maintenance of anesthesia via a nose mask with 1.5-2% isoflurane. The neck fur was carefully clipped on both the anterior and posterior sides. Subsequently, mice were placed in a sterile field in a ventrally recumbent position, and a 1.5 to 2 cm horizontal incision was made. The pump and biotelemetry transmitter device were then implanted subcutaneously. The pump catheter was trimmed to 0.5 cm for subcutaneous drug delivery, while the telemetry catheter was tunneled subcutaneously over the shoulder, with its tip positioned within the left carotid artery. Closure of incisions was performed using 5-0 sutures, after which the mice were gradually brought out of anesthesia. It is important to note that due to the complexity of the surgeries involved, mice used for behavioral studies were not implanted with a telemetry catheter or cranial window. However, these mice were individually housed to simulate the conditions of the telemetry cohort.

Blood pressure assessment and BPV induction

After recovery from surgery, mice were housed individually in standard mouse cages under the conditions described above and assigned to a control or experimental BPV group. During this recovery period (~7 days), both groups were infused with saline to determine baseline cardiovascular parameters. Blood pressure signals encompassing mean arterial pressure (MAP), systolic blood pressure (SBP), diastolic blood pressure (DBP), heart rate (HR), and pulse pressure (PP) (determined from the difference between the SBP and DBP) were continuously sampled at 250 Hz for 10 seconds and collected every 30 seconds, for 24 hours per day. Once a baseline MAP was established, the saline in the infusion pump (in the BPV group) was substituted with angiotensin II (Ang II, Sigma Aldrich, A9525) and administered intermittently at a calculated dose of 3.1 µg/hour (every 3-4 hours) for 25 days.

Controls receiving Ang II

To facilitate between-group comparisons (control vs BPV), a cohort of control mice were subjected to the same pump infusion parameters as BPV mice but for shorter durations receiving Ang II infusions on days 3-5 and, then again, on days 21-25 of the experimental protocol. This paradigm was used for experiments assessing pressure-evoked responses, including bradycardic reflex, myogenic response, and functional hyperemia at high BP.

BPV induction with norepinephrine

To validate the intermittent infusion protocol as an effective method for inducing significant BPV and not limited to Ang II, a separate cohort of mice received norepinephrine (NE, 45 µg/kg/min) using the same subcutaneous delivery protocol (2 µL every 4 hours).

Two-photon imaging

On the day of two-photon imaging acquisition, a mouse was anesthetized with chlorprothixene (0.04 cc) and a low dose of isoflurane ($\leq 0.8\%$); a protocol to mildly sedate mice²⁹. A retro-orbital injection of Texas red (70kDa dextran, 5% [wt/vol] in saline, 40 µl) was administered to label blood vessels. The mouse was head-fixed and a picospritzer (positioned on the contralateral hemisphere intended for imaging) was used to deliver a puff of air for whisker stimulation (WS) at a rate of 10 Hz for 20 seconds repeated six to eight times with ~ 90-second delay between each WS during an imaging session. Imaging sessions were conducted 106 ± 3.6 µm below the brain's surface while pump infusion was *off* (low BP period) and again during Ang II infusions (high BP period). Images were collected at a rate of 3.75 frames/second.

Behavioral studies

Behavioral tasks were performed by an experimenter at the AU small animal behavioral core (SABC), who was blinded to the experimental protocol. BPV mice were subjected to behavioral tests at two time points. First, during the saline infusion period, with observations considered as baseline, and second, 25-26 days post-Ang II infusions. About 30 minutes before training and testing, animals were brought to the testing room to acclimate. Animals remained in the laboratory for 15 minutes following study completion. To eliminate olfactory cues, animal droppings were removed, and the space and objects were cleaned (dilute 50% (vol/vol) ethanol solution) between sessions where appropriate.

Novel object recognition (NOR) task

Animal behavior was assessed using the previously described NOR task procedure³⁰. Animals were acclimated and then familiarized with an opaque plastic chamber (78.7 cm x 39.4 cm x 31.7 cm) containing bedding for 10 minutes. The next day, a training session involved animals exploring two identical objects for 10 minutes before returning to their home cages. Object recognition was assessed one hour later by placing an animal in the NOR apparatus with a familiar object and a new or novel object for 10 minutes. The objects' positions and roles (familiar or novel) were randomly assigned. Objects were positioned about 40 cm apart, 19.3 cm from the two short walls and 19.3 cm from the two long walls of the chamber. Exploration of an object was defined as direct interaction with nostrils or head position towards the object at a distance ≤ 2 cm. For data inclusion, a mouse had to explore an object for at least 4 seconds and spend a minimum of 10 seconds of total object exploration. Recognition index was calculated using the formula:

$Recognition\ Index = \frac{(time\ spent\ at\ novel\ object)}{(time\ spent\ at\ novel + familiar\ objects)}$. The discrimination index was

calculated as: $Discrimination\ Ratio = \frac{(time\ spent\ at\ novel - familiar\ object)}{(time\ spent\ at\ novel + familiar\ objects)}$. The one-hour delay

for testing was determined based on earlier NOR tests assessing the cohort's recognition capacity at one, four, and 24-hour delays. The animals exhibited normal recognition capacity with a one-hour delay but failed at 4- and 24-hour delays. Hence, the one-hour delay was selected for subsequent tests.

Y-maze

Y-maze tasks were performed twenty-four hours after NOR tasks. The Y-maze consisted of three arms (35.4 cm long, 9.9 cm wide with a height of 13.8 cm). Spontaneous alternation behavior was assessed by randomly placing mice in one of the three arms and allowed to explore for 10 minutes. Arm entries were visually scored into a series where a triplet set of arm entries constituted an alternation. An alternation was defined as successive consecutive entries into three different arms, and maximum alternations as the total number of arm entries minus 2. Percent alternation was calculated as the proportion of true alternations out of maximum alternations ((# of true alternations/# of maximum alternations) × 100).

Data analysis

Two-day averaged values of 24-hour or 12-hour (active and inactive period) BP variables were compared to a baseline phase corresponding to saline infusion (~ 5 days). BPV was calculated using the average real variability (ARV) index ($ARV = \frac{1}{N-1} \sum_{K=1}^{N-1} |BP_{K+1} - BP_K|$) where K is the order of measurements and N denotes the number of BP readings. Absolute BP values were extracted every 15 minutes and averaged for the hour. The coefficient of variation (CV) was also used to calculate BPV. CV was calculated as the ratio of the standard deviation of hourly BP averages to the 24-hour BP average ($CV = \frac{\text{Standard Deviation (Hourly BP Average)}}{24 \text{ Hour BP Average}}$). Similar to BP variables, two-day averages of 24-hour and 12-hour ARV and CV were compared to the average value of the last 5 days during the baseline phase. Parenchymal arteriole diameter responses to changes in MAP (myogenic responses) and whisker stimulation were analyzed using MatLab and ImageJ^{31,32}. Simple linear regression determined from the relationship of SBP and HR assessed bradycardic responses; MAP and arteriole diameter changes assessed myogenic reactivity. For neurovascular coupling experiments, the baseline was defined as 30 frames before stimulus onset. The stimulus-response comprised 100 frames encompassing the 20-second stimulus and the first 6 seconds post-stimulus. Six to eight WS events were averaged for each run and responses were normalized to the baseline average.

Exclusion Criteria

Mice that died prematurely (before treatment onset) or exhibited abnormally elevated or low blood pressures following telemetry surgery, likely due to surgical complications, were excluded from the study.

During two-photon imaging sessions, we observed unexplained, random transient dilatory events (**Supplemental Figure 1B, D**) occurring in both control and BPV mice across low- and high-BP periods. Because similar random dilations have been documented in other mouse strains under comparable *in vivo* imaging conditions, these events may be attributed to the effects of the sedative. To isolate pressure-evoked parenchymal arteriole responses in the myogenic studies (**Figure 5**), transient dilatory events associated with baseline tone were excluded from the analysis, **Supplemental Figure 1F**.

Study Limitations

Experiments involving control and BPV mice were conducted across different seasons, raising the possibility for seasonal influences^{45,46}. To account for this, seven additional control mice were included to the original thirteen control mice, **Supplemental Figure 2**. Six of these mice underwent identical treatment but were allocated to a separate branch of the study and did not receive chronic cranial window implantation—a procedure that does not significantly affect BP, **Supplemental Figure 2A**. No differences in 24-hour averaged MAP were observed between control mice when grouped categorized by Georgia's seasonal climate Fall-Winter (October-April)

vs Spring-Summer (May-September), **Supplemental Figure 2B** [↗](#). Given the absence of seasonal effects on BP and the fact that mice were sourced from two independent suppliers (Jackson Laboratory and the National Institute of Aging Aged Colonies), we attribute our observations primarily to treatment effects.

GraphPad Prism 10 software (GraphPad Software, La Jolla, CA) was used for all statistical analyses. Values are expressed as mean $\pm$ SEM. A minimum of three mice was used for each experimental data set, and the specific sample size (n) is defined in figure legends. Data was tested for normal distribution, and statistical tests were used accordingly. Differences between two means within groups were determined using paired Student's *t*-test. Differences between groups were determined using Student's unpaired *t*-test or two-way ANOVA with corresponding multiple comparison *post hoc* test specified in figure legends. Statistical significance was tested at a 95% ($P < 0.05$) confidence level denoted with the corresponding symbol in figure legends.

Results

Pulsatile Ang II infusion-induced changes in blood pressure

We developed a novel murine model of high BPV using the experimental design shown in **Figure 1A** [↗](#). The protocol consisted of a baseline phase (~5 days), corresponding to saline infusions in all mice, followed by a treatment phase (~20-25 days), where mice were either left on saline (control group) or subjected to robust BP transients induced via Ang II infusions (BPV group). Subcutaneous pumps were programmed to infuse 2 μ L 6-8 times per day, with infusion periods corresponding to 1 hour every 3-4 hours (depending on the number of pulses), **Figure 1B** [↗](#), right panel.

During the pump-off periods, BP corresponded to 97 ± 2.0 mmHg and 91 ± 1.9 mmHg for MAP, 113 ± 2.6 mmHg and 105 ± 2.6 mmHg for SBP, 80 ± 1.7 mmHg and 78 ± 1.7 mmHg for DBP, and 34 ± 1.3 mmHg and 27 ± 1.8 mmHg for PP, for mice with pumps infusing saline vs. Ang II, respectively (**Figure 1C** [↗](#)). During the pump-on period, saline infusion did not change BP. However, Ang II infusions evoked significant BP transients corresponding to $\Delta 37 \pm 3.9$ mmHg ($P < 0.0001$) for MAP, $\Delta 70 \pm 3.0$ mmHg ($P < 0.0001$) for SBP, $\Delta 29 \pm 4.1$ mmHg ($P < 0.0001$) for DBP, and $\Delta 28 \pm 3.0$ mmHg ($P < 0.0001$) for PP, **Figure 1C** [↗](#). Note that large dynamic activity-induced BP changes were apparent during both saline- and Ang II-infusions, **Figure 1B** [↗](#). Despite robust BP transients, 20 days of treatment minimally affected the 24-hour averages (vs Baseline) in all cardiovascular variables, **Figure 1D** [↗](#). The delta BP (Baseline vs Day 20) for MAP was -1 ± 2.0 mmHg and 6 ± 2.4 mmHg for control and BPV mice, respectively (**Figure 1D** [↗](#)). While the SBP in BPV mice significantly increased $\Delta 9 \pm 2.9$ mmHg ($P = 0.0088$), mice remained within the lower range of high BP (131 ± 4.4 mmHg). Additionally, prior to treatment, the baseline DBP in BPV was significantly lower than in control mice ($P = 0.0108$), **Figure 1D** [↗](#).

Blood pressure exhibits a circadian rhythm that is not discernible in minute-to-minute traces, (**Figure 1B** [↗](#)) but remains intact in both controls and BPV mice when analyzed in 1-hour averages, **Supplemental Figure 3C-D** [↗](#). Thus, cardiovascular variables were assessed during the active (dark) and inactive (light) phases of the mouse 12:12 hr cycle. Consistent with the unchanged 24-hour average for MAP, there were minimal changes in the averaged BP corresponding to the daily inactive and active periods for control and BPV mice, **Figure 2A-D** [↗](#). Compared to controls, the DBP of the BPV group tended to be lower but not significant ($P = 0.09$), **Figure 2C** [↗](#). However, beginning on day 7 of treatment, BPV (vs control) mice exhibited significantly higher PP during the inactive period, **Figure 2D** [↗](#). These data suggest a compensatory mechanism that maintains MAP within physiological ranges, albeit with elevated PP in BPV mice.

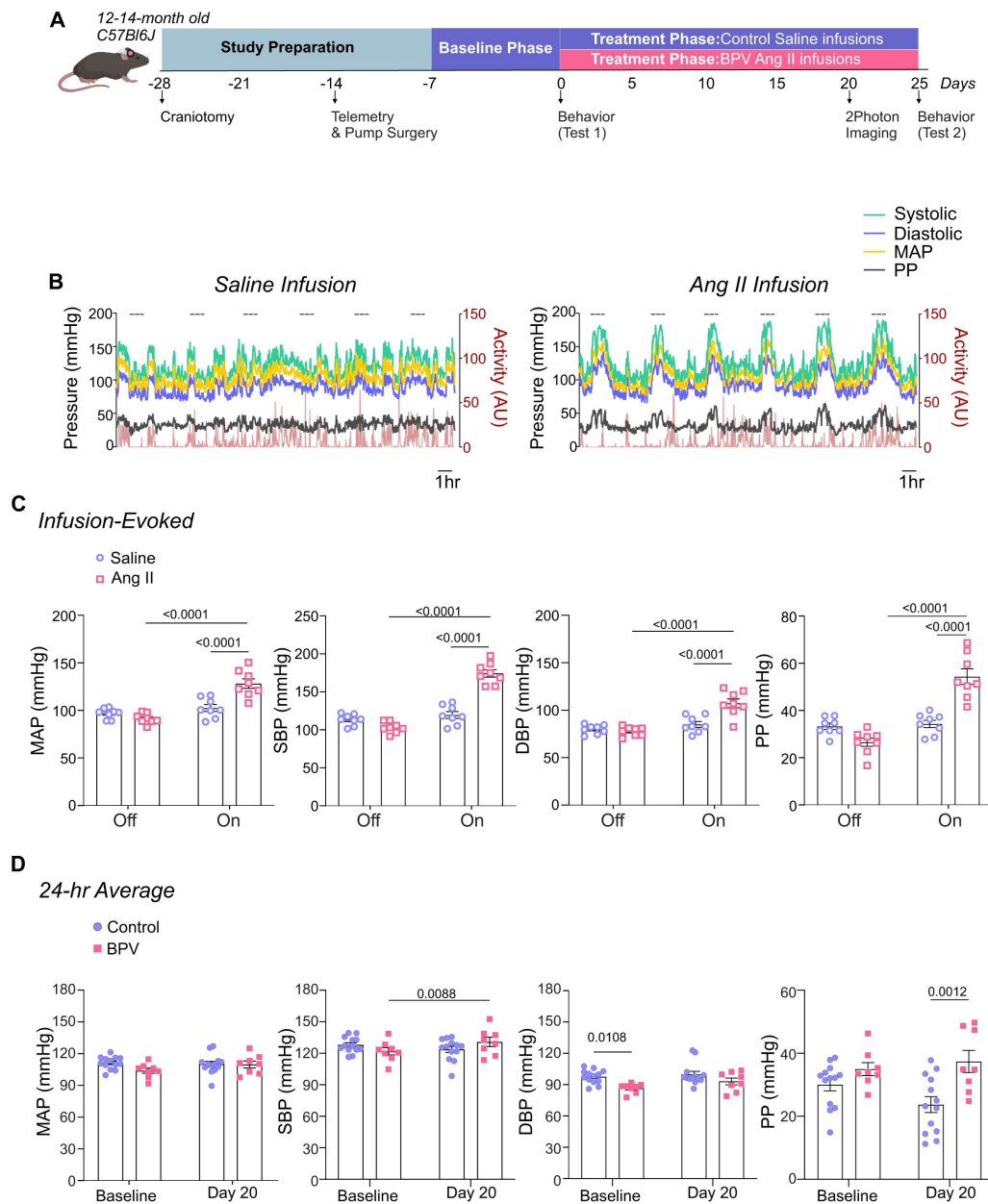


Figure 1.

Experimental design and Ang II infusion effects on mean arterial pressure.

A, Schematic of the experimental design for middle-aged C57BL/6J male mice, including implantation of a chronic cranial window (~28 days from treatment onset) and an infusion pump with telemetry (~7 days from treatment onset). Infusion pumps were programmed for intermittent delivery of saline (baseline phase) or angiotensin II (Ang II) (treatment phase) at a rate of 18.4 $\mu\text{g}/\text{day}$, administered in 1-hour intervals every 3-4 hours. *In-vivo* two-photon (2P) imaging sessions were performed approximately 20 days into the treatment phase. Behavioral tests were conducted during the baseline period (Test 1) and 25 days into the treatment phase (Test 2). This panel was created using *BioRender/v89f171*. **B**, Representative raw traces depicting minute-to-minute, 24-hour MAP, SBP, DBP, PP, and Activity levels during intermittent saline infusions (left) and Ang II infusions (right). Dashed lines indicate periods of active infusion (1 hr) or when the pump is *on*. **C**, Average MAP, SBP, DBP, and PP (5 minutes) measured during infusion *Off* versus *On* conditions. **D**, 24-hour average MAP, SBP, DBP, and PP during a 5-day saline infusion phase (Baseline) and on Day 20 of Ang II treatment. Two-way ANOVA repeated measures followed by Sidak's multiple comparisons test (C, $n = 8$ BPV mice) (D, $n = 13$ control mice, $n = 8$ BPV mice). **Ang II**=Angiotensin II, **BPV**=blood pressure variability, **DBP**=diastolic blood pressure **MAP**=mean arterial pressure, **PP**=pulse pressure, **SBP**=systolic blood pressure.

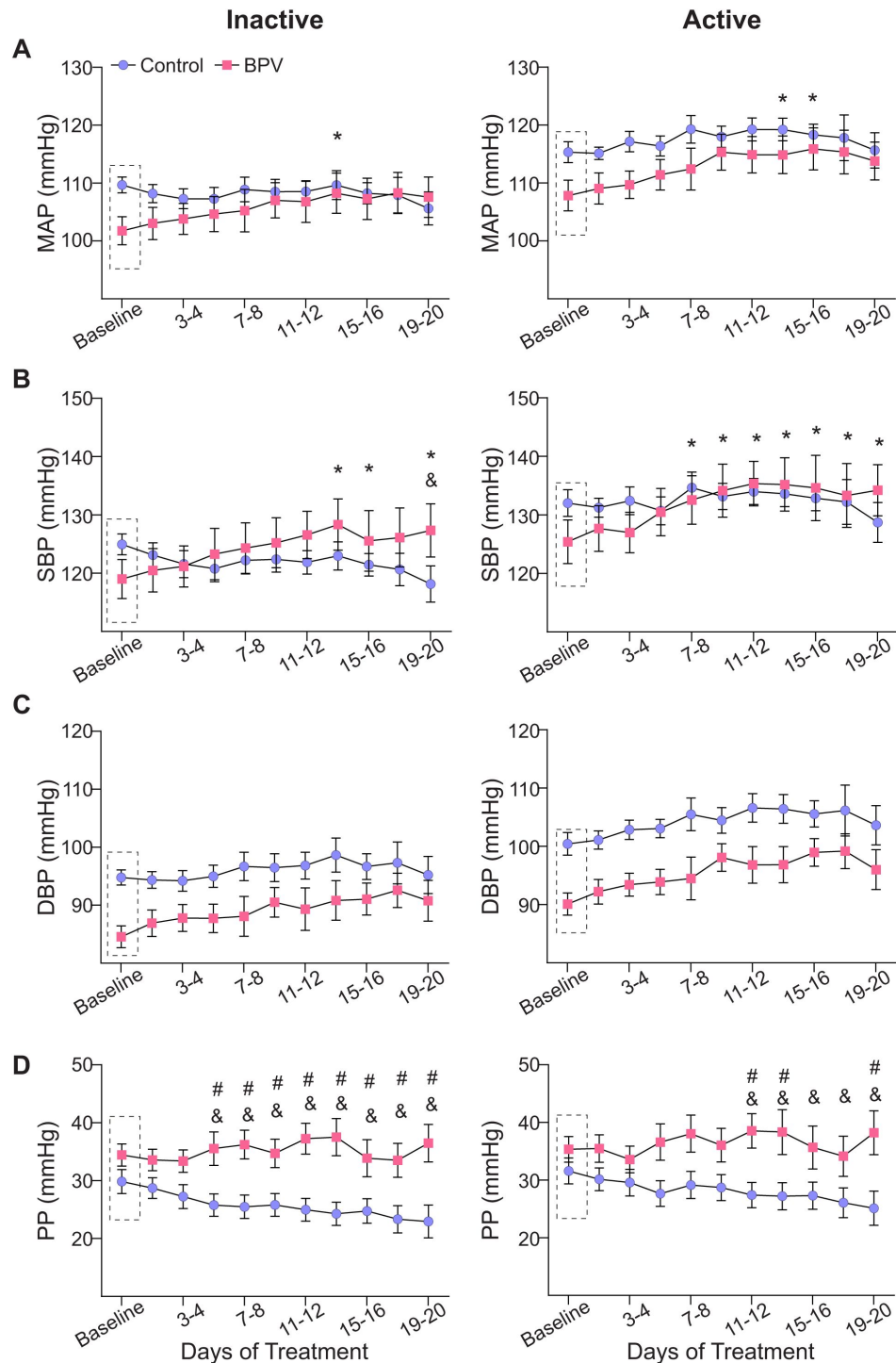


Figure 2.

Effects of pulsatile BP on average 12:12 hour cardiovascular variables.

A, Summary data of two-day averages of MAP, SBP (**B**), DBP (**C**), and PP (**D**) during the inactive (daylight) period (left column) and the active (nighttime) period (right column). Dashed rectangles indicate the baseline period (~5-day saline infusion). “*” and “&” denote within group comparisons ($p < 0.05$ vs Baseline) for BPV and control, respectively. “#” denotes $p < 0.05$ between groups comparisons. Two-way ANOVA repeated measures followed by Sidak’s comparison test; $n = 13$ control mice, $n = 8$ BPV mice.

BPV=blood pressure variability, **DBP**=diastolic blood pressure, **MAP**=mean arterial pressure, **PP**=pulse pressure, **SBP**=systolic blood pressure.

Pulsatile Ang II infusion induced high BPV

Having achieved minimal changes in the 24-hour average MAP, yet with prominent BP fluctuations, we quantified blood pressure variability (BPV) over time. Two indices of BPV were assessed: the average real variability (ARV), defined as the absolute difference between consecutive BP measurements, and the coefficient of variation (CV), defined as the ratio of the standard deviation to mean BP. A significant increase in BPV was observed in the BPV group over time relative to baseline, despite stable 24-hour BP averages. During the active period, ARV was significantly increased by day 3 and remained increased throughout the protocol, reaching 16.4 ± 0.66 mmHg ($P=0.04$) for SBP and 8.12 ± 0.71 mmHg ($P=0.04$) for PP, [Table 1A](#). Similarly, during the inactive period, ARV was significantly increased by day 3 and persisted, with values of 13.4 ± 0.33 mmHg ($P<0.01$) for MAP, 17.3 ± 0.48 mmHg ($P=0.02$) for SBP, 11.3 ± 0.47 mmHg ($P<0.01$) for DBP, and 7.9 ± 0.55 mmHg ($P=0.04$) for PP ([Table 1B](#)) with SBP as the main variable driving BPV increases in this model. BPV indices using CV, based on hourly BP averages over 24 hours, were significantly increased from day 1 of treatment across all cardiovascular variables during both the inactive and active periods, except for DBP of the active period, which increased on day 3, [Supplement Table 1](#).

Autonomic function in BPV mice

To compare pressure-induced bradycardic responses between BPV and control mice at both early and later treatment stages, a cohort of control mice received Ang II infusion on days 3-5 (early phase) ([Supplemental Figure 4](#)) and days 21-25 (late phase) thereby transiently increasing BP. Ang II evoked significant increases in SBP in both the control and BPV groups which were accompanied by a pronounced bradycardic response, [Figure 3A](#). For a single infusion-evoked BP peak, changes in HR were $-\Delta 291 \pm 34$ bpm ($P<0.0001$) and $-\Delta 304 \pm 17$ bpm ($P<0.0001$) for control and BPV mice, respectively ([Figure 3B](#)). While each Ang II infusion evoked significant bradycardic responses, the 24-hour HR average was not significantly altered, corresponding to 612 ± 7 and 585 ± 12 bpm for baseline and 570 ± 25 and 546 ± 8 bpm at day 20, for control and BPV mice, respectively, [Figure 3C](#).

To assess whether chronic BP fluctuations affect the baroreflex response, we compared the slopes extracted from linear regression of SBP vs HR, focusing on within-group differences, [Figure 3D-G](#). Comparisons were conducted over a 100-minute window surrounding a single Ang II-evoked BP pulse (20 minutes before and after 60-minute pulse), occurring during both the active and inactive periods, as well as between the early and late treatment phases. Linear regression analysis of SBP vs HR during Ang II infusion revealed no significant differences in the control group over time (Early vs Late), [Figure 3D, F](#). However, the BPV group exhibited a significant reduction in the bradycardic response, indicated by flatter negative slopes during both the inactive ($P<0.0001$) and active periods ($P<0.001$) ([Figure 3E, G](#)). These data suggest that chronic (~25 days) BPV diminishes bradycardic responses, indicative of suppressed autonomic function.

Pulsatile Norepinephrine infusion induced BPV

To determine whether the observed cardiovascular effects of BPV were limited to Ang II, norepinephrine (NE, 45 μ g/kg/min) was utilized as an alternative agent to induce BP variations, [Figure 4](#). The NE dosage was titrated to match Ang II-induced BP transient pulse levels (150-200 mmHg, SBP) using the same subcutaneous delivery protocol (pump on for 1 hour every 4 hours) resulting in 6 pulses/day, [Figure 4A](#). Similar to Ang II treated mice, NE-treated mice exhibited minimal changes in the 24-hr average MAP throughout the 20-day treatment, [Figure 4B](#). While there was a trend towards increased ARV ($P=0.08$), it did not reach significance, [Figure 4C](#) left panel. However, CV, another BPV index, was significantly increased by day 3 ($P=0.02$) of the treatment, [Figure 4C](#) right panel.

A

Active Period ARV mmHg $\pm$ SEM

Days	MAP		SBP		DBP		PP	
	Control	BPV	Control	BPV	Control	BPV	Control	BPV
Baseline	12.44 $\pm$ 1.37	10.94 $\pm$ 0.55	14.15 $\pm$ 1.49	13.13 $\pm$ 0.66	11.53 $\pm$ 1.44	9.04 $\pm$ 0.54	4.67 $\pm$ 0.40	5.47 $\pm$ 0.46
1-2	12.91 $\pm$ 1.29	10.99 $\pm$ 0.35	14.86 $\pm$ 1.28	14.56 $\pm$ 0.44	11.70 $\pm$ 1.48	9.29 $\pm$ 0.41	5.12 $\pm$ 0.40	7.67 $\pm$ 0.69
3-4	12.44 $\pm$ 0.66	12.51 $\pm$ 0.43	14.40 $\pm$ 0.67	16.40 $\pm$ 0.66 *	11.12 $\pm$ 0.83	10.35 $\pm$ 0.45	5.03 $\pm$ 0.33	8.12 $\pm$ 0.71 *#
5-6	11.78 $\pm$ 0.52	12.80 $\pm$ 0.39	13.64 $\pm$ 0.61	16.44 $\pm$ 0.53 *#	10.71 $\pm$ 0.73	10.54 $\pm$ 0.45	4.89 $\pm$ 0.32	7.99 $\pm$ 0.69 *#
7-8	12.35 $\pm$ 0.61	12.85 $\pm$ 0.41 *	14.02 $\pm$ 0.67	16.73 $\pm$ 0.43 *#	11.37 $\pm$ 0.81	10.61 $\pm$ 0.55 *	5.22 $\pm$ 0.42	8.34 $\pm$ 0.74 *#
9-10	12.34 $\pm$ 0.69	12.84 $\pm$ 0.40	13.88 $\pm$ 0.75	16.37 $\pm$ 0.72	11.32 $\pm$ 0.79	10.82 $\pm$ 0.24	4.95 $\pm$ 0.39	7.86 $\pm$ 0.65 *#
11-12	13.00 $\pm$ 0.76	12.94 $\pm$ 0.76 *	14.59 $\pm$ 0.88	16.87 $\pm$ 0.70 *	11.93 $\pm$ 0.93	10.59 $\pm$ 0.81	5.05 $\pm$ 0.34	8.25 $\pm$ 0.78 #
13-14	12.26 $\pm$ 0.84	12.90 $\pm$ 0.42 *	13.94 $\pm$ 0.84	16.80 $\pm$ 0.52 *	11.17 $\pm$ 0.95	10.40 $\pm$ 0.68 *	5.07 $\pm$ 0.39	8.45 $\pm$ 0.83 #
15-16	13.27 $\pm$ 0.59	13.91 $\pm$ 1.02	15.11 $\pm$ 0.81	17.50 $\pm$ 0.87 *	11.72 $\pm$ 0.82	11.59 $\pm$ 1.14	5.24 $\pm$ 0.38	8.25 $\pm$ 0.73 *
17-18	13.12 $\pm$ 0.77	14.06 $\pm$ 0.92	15.16 $\pm$ 1.03	17.12 $\pm$ 0.77 *	11.99 $\pm$ 0.85	11.95 $\pm$ 1.19	5.48 $\pm$ 0.58	7.60 $\pm$ 0.77
19-20	12.62 $\pm$ 0.45	12.75 $\pm$ 0.92	14.75 $\pm$ 0.52	16.57 $\pm$ 0.94	11.09 $\pm$ 0.70	10.48 $\pm$ 1.23	6.32 $\pm$ 0.60	8.44 $\pm$ 1.15

B

Inactive Period ARV mmHg $\pm$ SEM

Days	MAP		SBP		DBP		PP	
	Control	BPV	Control	BPV	Control	BPV	Control	BPV
Baseline	12.09 $\pm$ 1.38	11.29 $\pm$ 0.54	13.53 $\pm$ 1.50	13.28 $\pm$ 0.70	11.27 $\pm$ 1.40	9.48 $\pm$ 0.45	4.74 $\pm$ 0.47	5.04 $\pm$ 0.45
1-2	12.15 $\pm$ 1.04	11.40 $\pm$ 0.69	14.08 $\pm$ 1.03	14.69 $\pm$ 0.77	10.95 $\pm$ 1.15	9.76 $\pm$ 0.68	4.92 $\pm$ 0.24	7.09 $\pm$ 0.55
3-4	12.23 $\pm$ 0.79	13.38 $\pm$ 0.33 *	14.02 $\pm$ 0.76	17.27 $\pm$ 0.48 *#	11.19 $\pm$ 1.02	11.29 $\pm$ 0.47 *	4.76 $\pm$ 0.32	7.90 $\pm$ 0.55 *#
5-6	12.16 $\pm$ 0.43	13.63 $\pm$ 0.78 *	13.64 $\pm$ 0.52	17.57 $\pm$ 0.78 *#	10.68 $\pm$ 0.53	11.34 $\pm$ 0.86	4.95 $\pm$ 0.30	8.23 $\pm$ 0.69 *#
7-8	11.95 $\pm$ 0.49	13.73 $\pm$ 0.50 *	14.71 $\pm$ 0.54	17.44 $\pm$ 0.53 *#	10.88 $\pm$ 0.59	11.61 $\pm$ 0.67 *	4.90 $\pm$ 0.32	8.45 $\pm$ 0.89
9-10	12.90 $\pm$ 1.28	14.77 $\pm$ 0.66 *	14.09 $\pm$ 1.49	18.52 $\pm$ 0.72 *	11.92 $\pm$ 1.15	12.59 $\pm$ 0.64 *	5.33 $\pm$ 0.31	8.16 $\pm$ 0.61 *#
11-12	12.49 $\pm$ 0.79	14.53 $\pm$ 0.57 *	14.85 $\pm$ 0.82	18.65 $\pm$ 0.62 *#	11.50 $\pm$ 1.07	12.09 $\pm$ 0.74 *	5.24 $\pm$ 0.31	8.50 $\pm$ 0.88
13-14	13.03 $\pm$ 0.75	13.96 $\pm$ 0.74 *	14.85 $\pm$ 0.82	17.65 $\pm$ 0.53 *	11.94 $\pm$ 0.99	11.56 $\pm$ 0.92	5.10 $\pm$ 0.38	8.16 $\pm$ 0.74 *#
15-16	13.35 $\pm$ 0.79	15.04 $\pm$ 0.87 *	14.87 $\pm$ 0.93	18.85 $\pm$ 0.87 *	12.21 $\pm$ 0.98	12.66 $\pm$ 1.07	4.97 $\pm$ 0.36	8.54 $\pm$ 1.06 *
17-18	12.99 $\pm$ 0.70	15.61 $\pm$ 1.36	14.19 $\pm$ 0.82	19.39 $\pm$ 1.09 *#	11.47 $\pm$ 0.83	13.18 $\pm$ 1.67	5.02 $\pm$ 0.40	8.46 $\pm$ 1.03
19-20	12.89 $\pm$ 0.57	14.72 $\pm$ 1.00 *	15.13 $\pm$ 0.74	18.96 $\pm$ 0.96 *	11.36 $\pm$ 0.65	12.19 $\pm$ 1.25	6.21 $\pm$ 0.40	9.28 $\pm$ 1.33

Table 1.

Ang II-induced increases in blood pressure variability (BPV).

A-B, Summary data of calculated two-day average real variability of MAP, SBP, DBP, and PP during the active period (**A**) and the inactive period (**B**). Baseline corresponds to (~5-day saline infusion) and “*” denotes within group comparisons ($p < 0.05$ vs Baseline) and “#” denotes $p < 0.05$ between groups comparisons. Two-way ANOVA repeated measures followed by Dunnett’s comparison test; $n = 13$ control mice, $n = 8$ BPV mice.

ARV=average real variability, **BPV**=blood pressure variability, **DBP**=diastolic blood pressure, **MAP**= mean arterial pressure, **PP**=pulse pressure, **SBP**=systolic blood pressure.

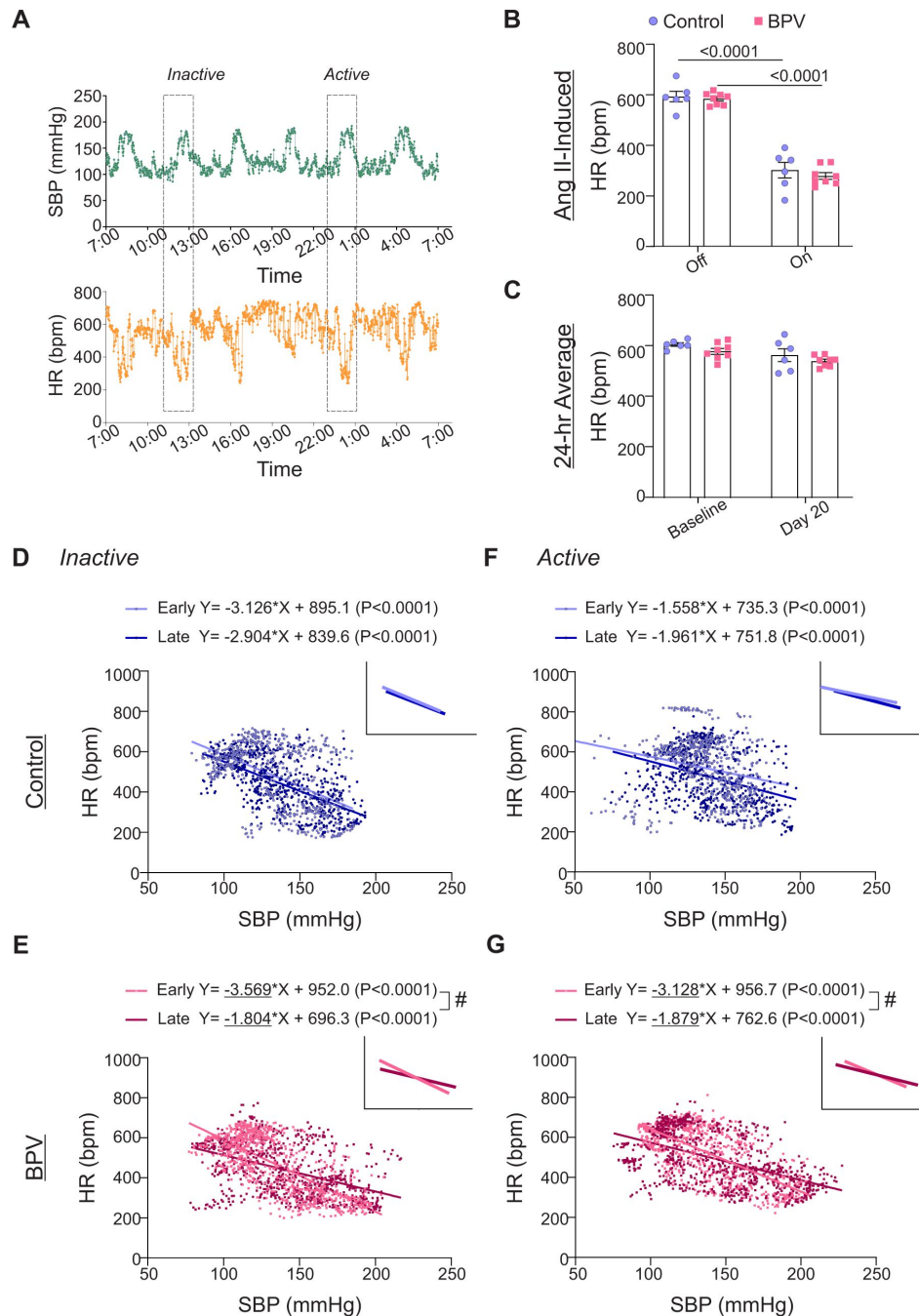


Figure 3.

Chronic BPV suppresses bradycardic reflex.

A, Representative raw traces of 24-hour SBP (top) and HR (bottom) during intermittent Ang II infusions. **B**, Five-minute average HR when the infusion pump is *Off* or *On* (actively infusing Ang II) for control and BPV groups, extracted from the inactive (daytime) period. **C**, Twenty-four-hour average HR during the baseline period (~5-day saline infusion), and on day 20 of treatment (Day 20) for control and BPV mice. **D**, Scatter plot of SBP and HR during Ang II infusion (100 minutes: 20 minutes before and after the 60-minute pulse) in the inactive period for control and BPV mice (**E**). Linear trend lines for data shown in the insert. Early and Late periods correspond to days 3-5 and days 23-25 of the treatment phase, respectively. **F**, Scatter plot of SBP and HR during the active period in controls and in BPV mice (**G**). Two-way ANOVA repeated measures followed by Sidak's multiple comparisons test (B-C, $n = 6$ control mice, $n = 8$ BPV mice). Simple linear regression and “#” denotes $p < 0.05$ within group (D, F, $n = 6$ control mice) (E, G, $n = 8$ BPV mice).

bpm = beats per minute, **BPV** = blood pressure variability, **HR** = heart rate, **SBP** = systolic blood pressure.

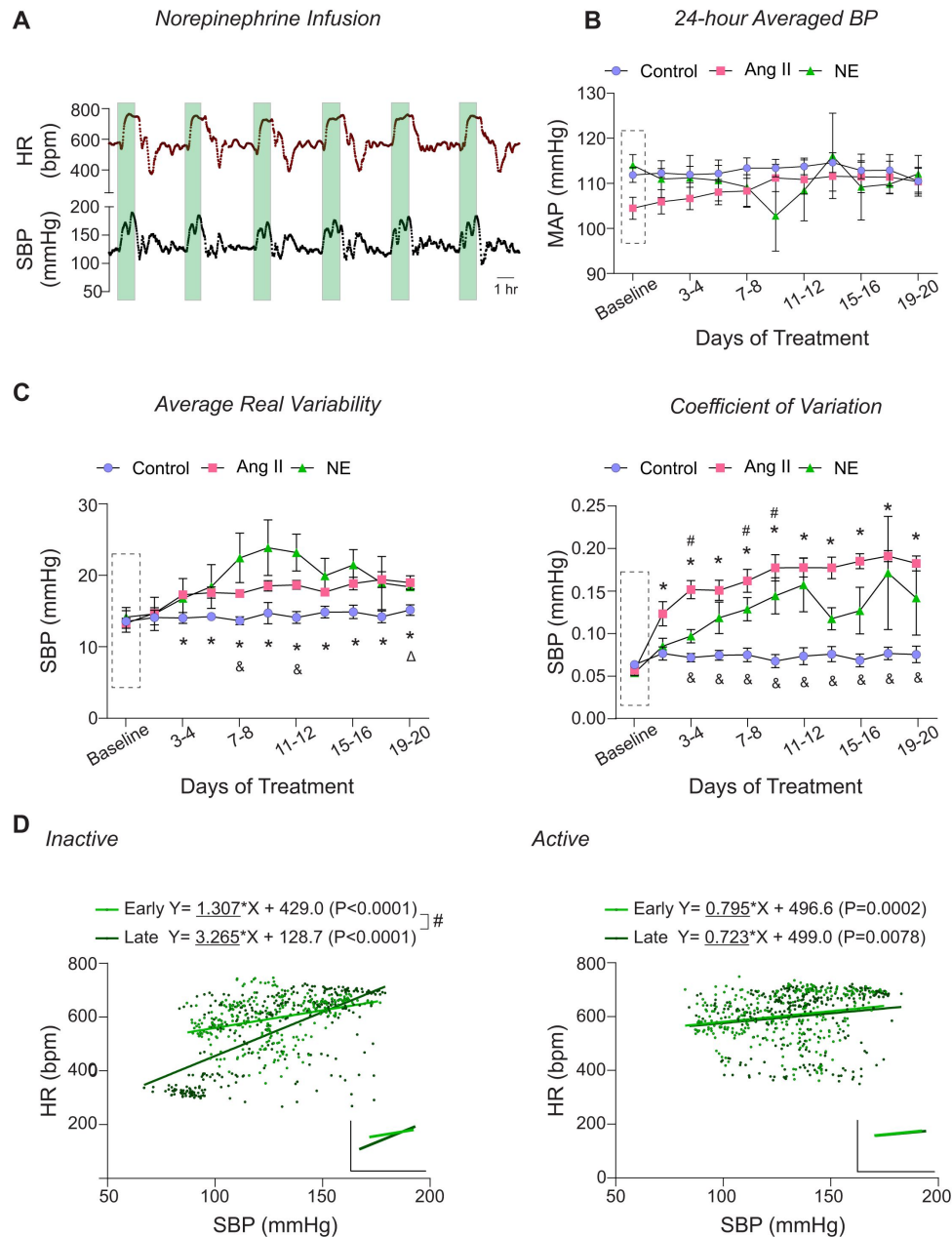


Figure 4.

Norepinephrine infusion effects.

A, Representative raw trace of minute-to-minute 24hr HR (top) and SBP (bottom). The green-shaded region indicates the 1-hour period when the pump is actively infusing NE (45 $\mu\text{g/kg/min}$). This panel was created using *BioRender/m19t088*. **B**, Summary data of two-day average MAP during the baseline phase (Dashed rectangle, ~5-day saline infusion) and over 20 days of treatment. **C**, Summary data of calculated two-day ARV of SBP (left) and CV of SBP (right). **D**, Scatter plot of SBP and HR during NE infusion (100-minute window: 20 minutes before and after the 60-minute pulse) in the inactive (left) and active (right) periods, with linear trend lines shown in the insert. Early and Late periods correspond to days 3-5 and days 16-20 of the treatment phase. “*” and “#” denote within group comparisons ($p < 0.05$ vs Baseline) for Ang II- and NE-infused mice, respectively. “&” and “Δ” denotes between group comparisons ($p < 0.05$ vs control group) for Ang II- and NE-infused mice, respectively. Two-way ANOVA repeated measures followed by Sidak’s multiple comparisons test (B-C, $n = 6$ mice). Simple linear regression (D, $n = 3$ mice).

Ang II=Angiotensin II, **bpm** = beats per minute, **HR** = heart rate, **NE** = norepinephrine, **SBP** = systolic blood pressure.

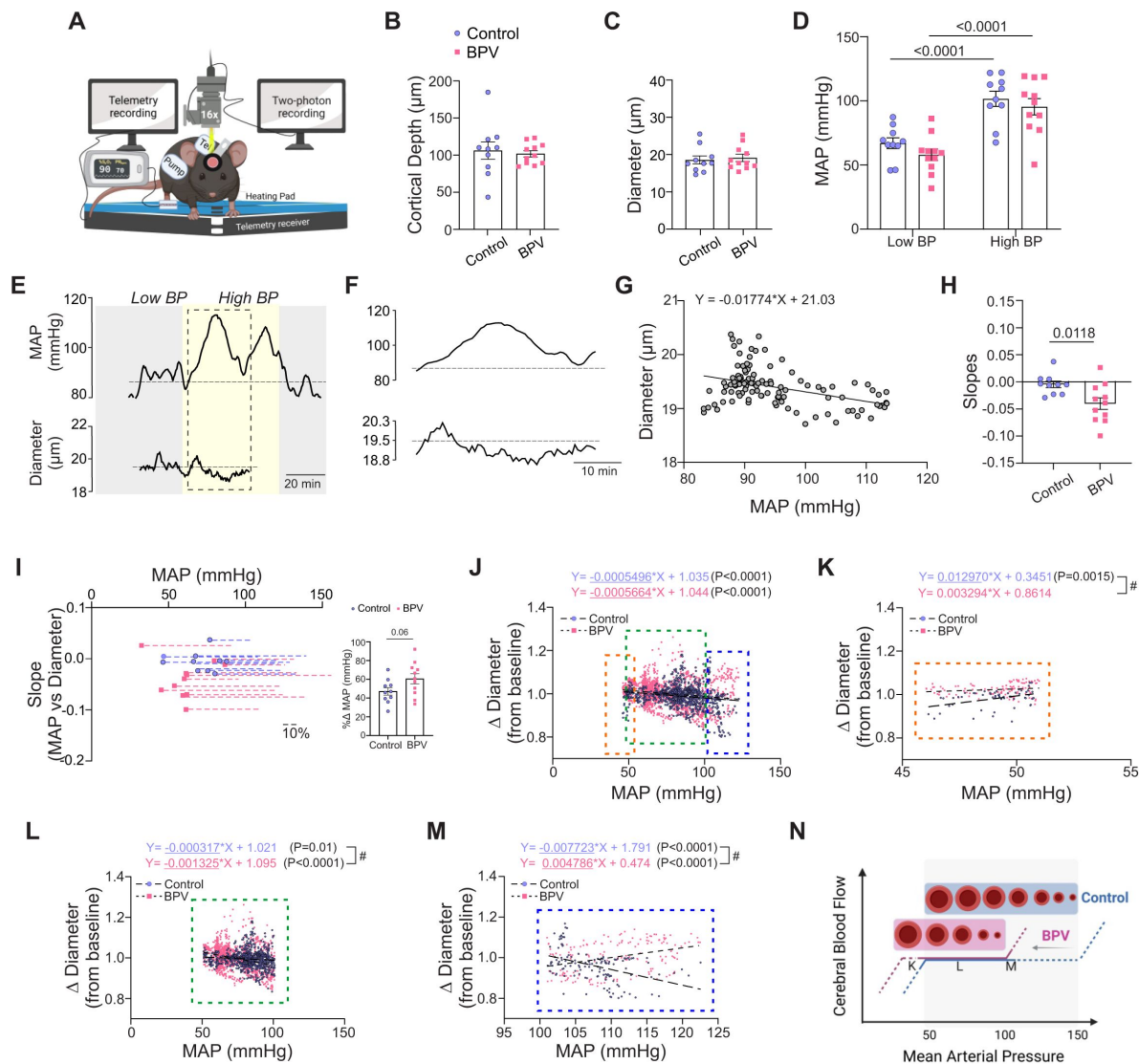


Figure 5.

Enhanced myogenic responses in parenchymal arterioles of BPV mice.

A, Schematic of the experimental set-up, including a portable telemetry system for continuous blood pressure recordings alongside simultaneous 2P imaging of parenchymal arteriole diameter. This panel was created using *BioRender/c21h358*. **B**, Summary data for the average imaging depth below the brain surface. **C**, Summary data for the average baseline diameters of imaged parenchymal arterioles. **D**, Summary data of the average MAP recorded under infusion pump *off* conditions (low blood pressure, Low BP) and infusion pump *on* conditions (high blood pressure, High BP). **E**, Representative raw trace (from a single mouse) showing MAP (top) and parenchymal arteriole diameter (bottom) over time. **F**, Expanded data corresponding to dashed region in **(E)**, highlighting Ang II-evoked BP changes. **G**, Representative scatter plot and linear regression of MAP vs parenchymal arteriole diameter during an imaging session. **H**, Summary data of MAP-diameter linear regression slopes. **I**, Relationship between minimum MAP recorded during an imaging run and the slope shown in **(H)**; dashed lines indicate the percent Δ change in MAP relative to its minimum value. The insert provides a summary of percent Δ change in MAP from the minimum MAP. **J**, Scatter plot of MAP vs changes in arteriole diameter from baseline pressure (~ 70 mmHg), spanning a pressure range 46-122 mmHg. Data subsets extracted from **(J)**: **(K)** MAP ≤ 51 mmHg (orange dashed region), **(L)** MAP 51-100 mmHg (green dashed region), and **(M)** MAP ≥ 101 mmHg (blue dashed region). **N**, Illustration of the proposed leftward shift (and narrower plateau) in cerebral autoregulatory curve induced by chronic BPV. Unpaired t-test and Mann-Whitney test (B-C, H-I, $n = 10$ runs/8 control mice, $n = 11$ runs/8 BPV mice). Two-way ANOVA repeated measures followed by Sidak's multiple comparison test (D, $n = 10$ runs/8 control mice, $n = 11$ runs/8 BPV mice). Simple linear regression and "#" denotes $p < 0.05$ between group (J-M, $n = 12$ runs/9 control mice, $n = 8$ runs/7 BPV mice). **BPV** = blood pressure variability, **MAP** = mean arterial pressure

In contrast to the bradycardic response observed during Ang II-induced BP elevations, NE-induced BP pulses triggered a tachycardic response (**Figure 4A** [↗](#)), with significant positive correlation between SBP and HR appearing during the early phase in the treatment (Inactive period, $P < 0.0001$; Active period, $P = 0.0002$), **Figure 4D** [↗](#). These data validate the intermittent infusion protocol as an effective method for inducing significant BPV while preserving normal 24hr MAP. However, due to fundamental differences in bradycardic responses and the presence of off-target effects, the physiological actions of this vasopressor agent are not directly comparable to those of Ang II. These results underscore the necessity of careful vasopressor selection to mitigate off-target effects.

Chronic BPV enhanced parenchymal arteriole myogenic responses

CBF regulation involves complex processes that maintain the metabolic needs of working neurons. These include mechanisms active at baseline (e.g., cerebral autoregulation¹⁶ [↗](#) and endothelial cell-mediated signaling³³ [↗](#)) and those involved in activity-mediated increases in CBF or neurovascular coupling (NVC)¹⁴ [↗](#). To assess if BPV impacts the functional integrity of the neurovascular complex, we measured parenchymal arteriole responses to acute BP increases and determined if BP itself altered the sensory-evoked arteriole response. For these studies, *in vivo* two-photon imaging was combined with BP measurements via telemetry, **Figures 5** [↗](#) and **6** [↗](#).

To determine the impact of acute BP changes on the microcirculation, we simultaneously tracked parenchymal arteriole diameter changes before pump infusion onset (Low BP), and during Ang II infusions (High BP), **Figure 5E** [↗](#). For these experiments, and prior to the imaging session, the pump reservoir for the control group was switched from saline to Ang II. Imaging was conducted at similar cortical depths (~ layer 1/2 border of the somatosensory cortex) corresponding to $106 \pm 11.6 \mu\text{m}$ and $102 \pm 4.3 \mu\text{m}$ for control and BPV mice, respectively, **Figure 5B** [↗](#). Arteriole diameters measured at baseline (Low BP) were comparable between groups, corresponding to $18.6 \pm 1.1 \mu\text{m}$ and $19 \pm 1.0 \mu\text{m}$ for controls and BPV mice, respectively, **Figure 5C** [↗](#). Upon mild sedation resting BP dropped by $-23 \pm 3.6 \text{ mmHg}$ and $-31 \pm 4.4 \text{ mmHg}$ for the control and BPV group, respectively (not shown). However, BP values remained within the putative cerebral autoregulation plateau range throughout two-photon imaging sessions and were comparable between groups. During the pump infusion period (High BP), the BP increased from 68 ± 4.2 to $102 \pm 5.9 \text{ mmHg}$ for controls ($P < 0.0001$) and from 59 ± 4.5 to $96 \pm 6.4 \text{ mmHg}$ for BPV mice ($P < 0.0001$), **Figure 5D** [↗](#).

To quantify *in vivo* pressure-evoked diameter changes (myogenic responses), we compared the slopes extracted from the linear regression of MAP vs parenchymal arteriole diameter measured throughout the acquisition session, **Figure 5E-G** [↗](#). The steeper negative slope observed in BPV mice compared to controls indicates significantly increased parenchymal arteriole reactivity to BP changes ($P = 0.0118$), **Figure 5H** [↗](#). However, because BPV mice exhibited lower baseline BP, although not significantly different from controls ($P = 0.18$, **Figure 5D** [↗](#)), we examined whether differences in pressure range (i.e. the change in pressure between low and high BP intervals) contributed to the steeper slopes observed in BPV mice. **Figure 5I** [↗](#) illustrates the relationship between the lowest BP recorded during the acquisition period and the slope derived from the MAP-diameter linear regression analysis. Dashed lines represent the percent change in MAP relative to the minimal averaged BP recorded at the start of the imaging session. BPV mice showed a trend for greater percent change in BP ($\% \Delta 61 \pm 5.4 \text{ mmHg}$ vs $\% \Delta 47 \pm 4.0 \text{ mmHg}$ for control mice, $P = 0.06$; **Figure 5I** [↗](#), insert). However, the average BP range in control mice was still sufficient for establishing a correlation between BP and diameter, supporting the interpretation that differences in myogenic response stem from intrinsic vascular reactivity rather than limitations in the BP range.

The static cerebral autoregulation curve (CA), originally described by Lassen in 1959, is characterized by a plateau phase in the MAP-CBF relationship, sandwiched between lower and upper limit ranges³³ [↗](#). However, the shape of the curve has been challenged³⁴ [↗](#), with evidence

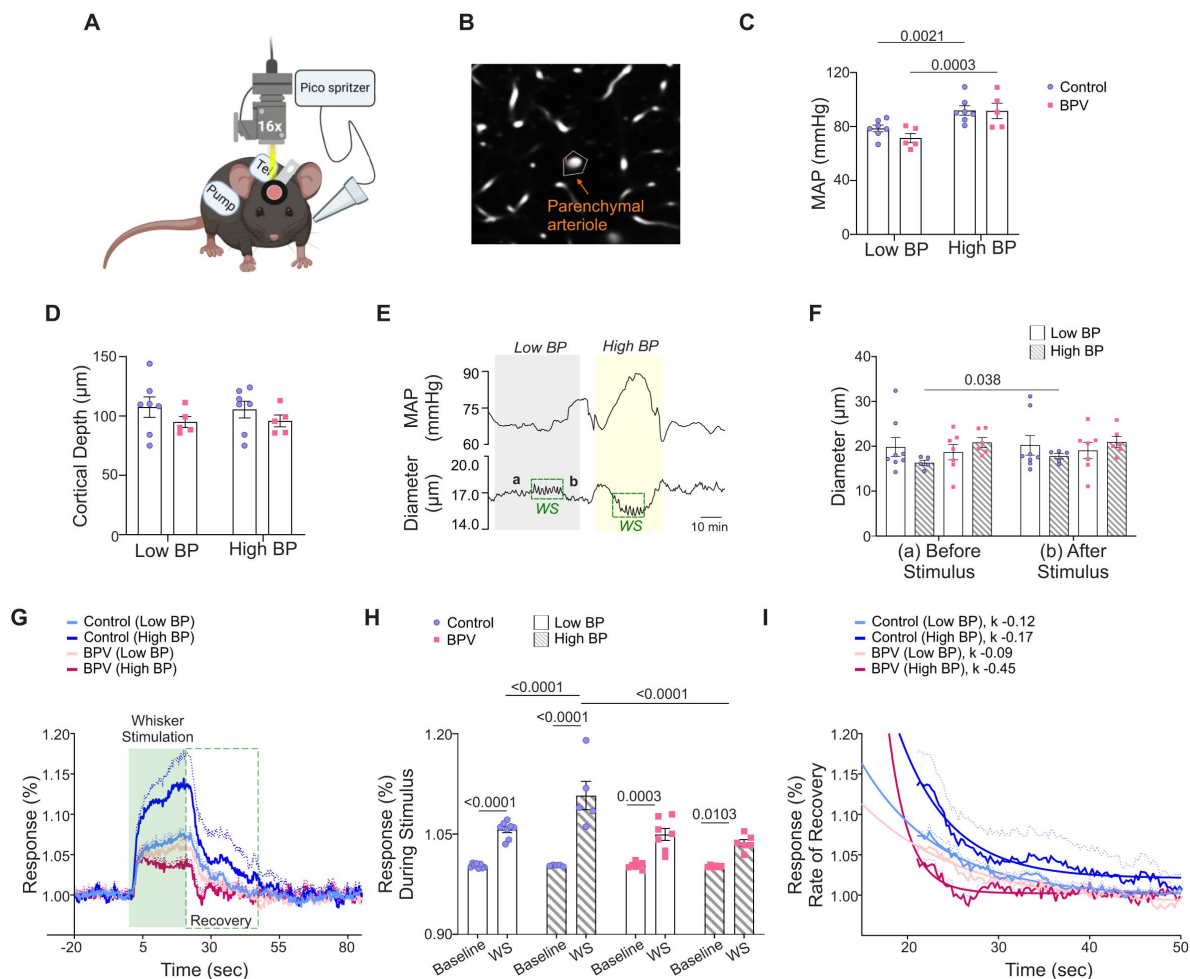


Figure 6.

Suppressed neurovascular responses in parenchymal arterioles of BPV during low and high blood pressure periods.

A, Schematic of the experimental setup in which a picospritzer delivered a puff of air for whisker stimulation (WS) at 10 Hz for 20 seconds. **B**, Representative image showing the mask (outlined) used to track changes in parenchymal arteriole diameter. Panels A and B were created using *BioRender/b34x028*. **C**, Summarized data of average MAP recorded during low blood pressure (Low BP; pump off) and high blood pressure (High BP; pump on) conditions. **D**, Summary data of the average imaging depth below the brain surface. **E**, Representative raw trace (from a single mouse) showing MAP (top) and parenchymal arteriole diameter (bottom) during low and high blood pressure. Dashed green squares indicate WS response, with "a" denoting the 30-second pre-stimulus diameter and "b" denoting the 30-second post-stimulus diameter, summarized in **(F)**. **G**, Normalized averaged arteriole diameter traces with corresponding error bars (dashed lines) during the WS response, shown as % change from baseline (20 seconds before stimulus), during the 20-second stimulus (green shaded region), and 64 seconds post-stimulus (green dashed square). **H**, Summarized data of stimulus-induced arteriole responses (green shaded region in **G**). **I**, Summary data of recovery time, with rate of decay (k) corresponding to 30-second post-stimulus period outlined in the green dashed square in **(G)**. Two-way ANOVA repeated measures followed by Sidak's multiple comparisons test (C-D, $n = 7$ control mice, 5 BPV mice) (F, $n = 5-8$ control mice, $n = 6-7$ BPV mice) and (H, $n = 5-8$ control mice, $n = 6-7$ BPV mice). One phase exponential decay nonlinear fit (I, $n = 5-8$ control mice, $n = 6-7$ BPV mice).

BPV = blood pressure variability, **k** = rate of decay, **MAP** = mean arterial pressure, **WS** = whisker stimulation

suggesting a narrower plateau range and potential dilatory phases as BP deviates from this range³⁵. To investigate pressure-diameter relationships in our cohort of mice, we analyzed parenchymal arterioles from control and BPV mice. As shown in **Figure 5J**, arterioles of both groups exhibited significant increases in tone ($P < 0.0001$) as pressure rose from ~50 to 125 mmHg. When slopes from segmented regression analyses were compared across autoregulatory limits and plateau ranges³⁵, our data revealed a significant decrease in arteriole diameter ($P = 0.0015$) in control mice at pressures < 50 mmHg (**Figure 5K**). Within the putative CA plateau range (51–100 mmHg)³⁵, arterioles from both groups showed significant constrictions ($P = 0.01$ for controls; $P < 0.0001$ for BPV mice, **Figure 5L**), with steeper slopes observed in BPV arterioles ($P < 0.0001$). At pressures > 100 mmHg, BPV arterioles exhibited significant arterial dilations ($P < 0.001$), whereas control arterioles maintained constrictions ($P < 0.001$, **Figure 5M**). Together, these data suggest a leftward shift in the CA curve of BPV mice, characterized by enhanced myogenic constrictions at comparable pressures and significant arterial dilations at higher pressures where controls arterioles maintain tone, **Figure 5N**.

Previous studies have shown directional sensitivity in parenchymal arteriole responses, whereby increases in MAP evoke more efficient responses as compared to decreases in MAP. Thus, we separated and quantified diameter changes evoked by BP transitions from Low-to-High (increase MAP) and High-to-Low BP (decrease MAP), **Supplemental Figure 5A–F**. Ang II infusion evoked similar increases in MAP when BP transitioned from Low-to-High $\Delta 27 \pm 3.6$ mmHg in controls and $\Delta 25 \pm 3.6$ mmHg in BPV mice, **Supplemental Figure 5A**. In addition, MAP decreases were comparable between groups ($-\Delta 21 \pm 2.8$ mmHg for controls and $-\Delta 25 \pm 3.1$ mmHg for BPV mice) upon cessation of pump infusion (High-to-Low), **Supplemental Figure 5D**. The significantly steeper negative slopes in BPV arterioles relative to controls ($P = 0.0117$, Low-to-High) (**Supplemental Figure 5C**) demonstrate increased parenchymal arteriole constriction to increases in MAP. Conversely, we also observed steeper negative slopes in BPV arterioles compared to controls ($P = 0.0044$, High-to-Low) (**Supplemental Figure 5F**) with decreases in MAP suggestive of enhanced arteriole dilations. Collectively, these findings indicate that chronic BPV heightens the responsiveness of parenchymal arterioles to acute BP fluctuations.

Impaired NVC in high BPV mice

Because mechanisms underlying baseline CBF regulation differ from those evoked during the functional hyperemia response, we assessed whether chronic BPV affected functional-evoked outputs or neurovascular coupling (NVC). Functional hyperemia was assessed using two-photon imaging and evoked using air puffs to stimulate mouse whiskers. Blood pressure measurements were simultaneously measured via telemetry, **Figure 6A**. While sensory-evoked vascular responses are commonly used as a proxy for neuronal activation and function^{18,34}, few studies have considered (or reported) BP during these *in vivo* protocols. Thus, we asked if chronic and acute BP fluctuations alter the NVC response at the level of parenchymal arterioles, **Figure 6B**. NVC responses were compared during low and high BP periods corresponding to pumps being *off* or *on*, respectively, **Figure 6E**.

The average baseline (low BP) MAP was 78 ± 2.5 mmHg and 71.0 ± 3.4 for the control and BPV group, respectively, **Figure 6C**. Imaging was conducted at similar cortical depths within ~ layers 1/2 border of the somatosensory cortex, corresponding to 107 ± 1.0 μ m and 96 ± 0.4 μ m for control and BPV arterioles, respectively, **Figure 6D**. To compare NVC responses at low and high BP, the pump reservoir for the control group was switched from saline to Ang II. Ang II infusion evoked significant (and comparable) increases in MAP, corresponding to 92 ± 3.5 mmHg for controls and 92 ± 5.7 mmHg for BPV mice, **Figure 6C**. **Figure 6E**, show a representative trace of the simultaneous MAP and arteriole diameter changes; WS denotes the stimulation period and “a” and “b” the timepoint from where diameters were measured before and after the stimulus, respectively and summarized in **Figure 6F**. Whisker stimulation caused significant dilations (vs. baseline) in all groups, regardless of MAP, **Figure 6G, H**. However, at higher BP, the magnitude of the NVC response in control mice was greater ($P < 0.0001$, high BP vs low BP). Notably, this

pressure-dependent response was abrogated ($P=0.60$) in the BPV group, **Figure 6G, H**. A closer look at the post-stimulus recovery phase showed a faster recovery rate ($k = -0.45$) for parenchymal arterioles of BPV mice during high BP periods compared to controls ($P<0.0001$), **Figure 6I**. These data revealed an acute pressure-dependent effect in control mice, with higher pressures resulting in greater sensory-evoked dilations. Additionally, the data shows that chronic BP fluctuations abrogated the pressure-dependent effects observed in control mice.

Cognitive decline in high BPV mice

Given impaired neurovascular outputs and the established association between high BPV and cognitive decline³⁵, we used the NOR and Y-maze tests to assess cognitive performance. Mice were subjected to behavioral testing during the saline-infused period and following 25 days of BPV (via Ang II infusions). Using NOR, a significant decrease in the recognition index ($P=0.006$) and discriminatory index ($P=0.006$) was observed (**Figure 7A**), supporting impaired recognition memory. No differences were observed in the short-term spatial working memory assessed via Y-maze alternation ($P=0.25$), **Figure 7B**.

Mice activity levels were evaluated as another aspect of behavior. Throughout the study, activity remained consistently higher during the active vs inactive periods ($P=0.049$ for control, $P=0.0287$ for BPV), **Figure 7C**. However, following 23-25 days of treatment (Late), BPV mice exhibited a significant reduction in activity when the pump was *on*. This decline was evident in the 24-hour average data ($P=0.0111$, **Figure 7D**), as well as during the active period of the 12-hour activity averages (pump *on* vs *off*, $P=0.0122$, **Figure 7F**). These data support that mice subjected to chronic BPV become less active when the BP increases or the pump status is *on*. Notably, while the infusion protocol preserves BP-related circadian rhythms, 25 days of chronic BPV negatively impacts both recognition memory and overall activity levels.

Discussion

BPV has emerged as a risk factor for cognitive decline, yet its impact on brain function remains poorly understood. Here, we introduce a novel murine model of BPV, where pulsatile BP increases were induced via Ang II infusions without hypertension. Using *in vivo* two-photon imaging, we provide evidence of altered vascular function in mice subjected to chronic BP fluctuations. In BPV mice, parenchymal arterioles ($<30\ \mu\text{m}$ in diameter) exhibited enhanced myogenic reactivity. In addition, chronic BPV impaired neurovascular outputs and cognitive function. Notably, the NVC response in control mice was pressure-dependent, with greater magnitudes observed when the arterial BP was increased—a relationship that was abolished in mice exposed to chronic BPV. These findings suggest that chronic BPV targets multiple physiological processes (e.g., bradycardic reflex, myogenic reactivity, neurovascular coupling) reinforcing its role as a critical risk factor for brain health.

Our goal was to induce rapid transient blood pressure pulses that significantly increased BPV, while maintaining a stable 24hr average blood pressure. Ang II was selected as a pressor due to its potent vasoconstrictive properties and short half-life³⁶. The subcutaneous dose was carefully titrated to induce BP elevations reaching physiological high levels (150–200 mmHg, SBP), achievable in non-diseased conditions and comparable to doses used in miniosmotic pumps^{37–39}. The precise mechanisms underlying increased BPV remain unclear, though vascular dysfunction and arterial stiffness have been postulated as contributing factors⁴⁰. Age is strongly correlated with arterial stiffness and isolated systolic hypertension⁴¹. Notably, SBP was modestly increased downstream of BPV in the current model, **Figure 1D**. In the BPV cohort, six out of eight mice exhibited an SBP increase of $>\Delta 10$ mmHg compared to baseline (**Figure 1D**), suggesting that elevated BPV may contribute to the early onset of systolic hypertension. Given that isolated systolic hypertension is a risk factor for overt hypertension⁴² and BPV precedes

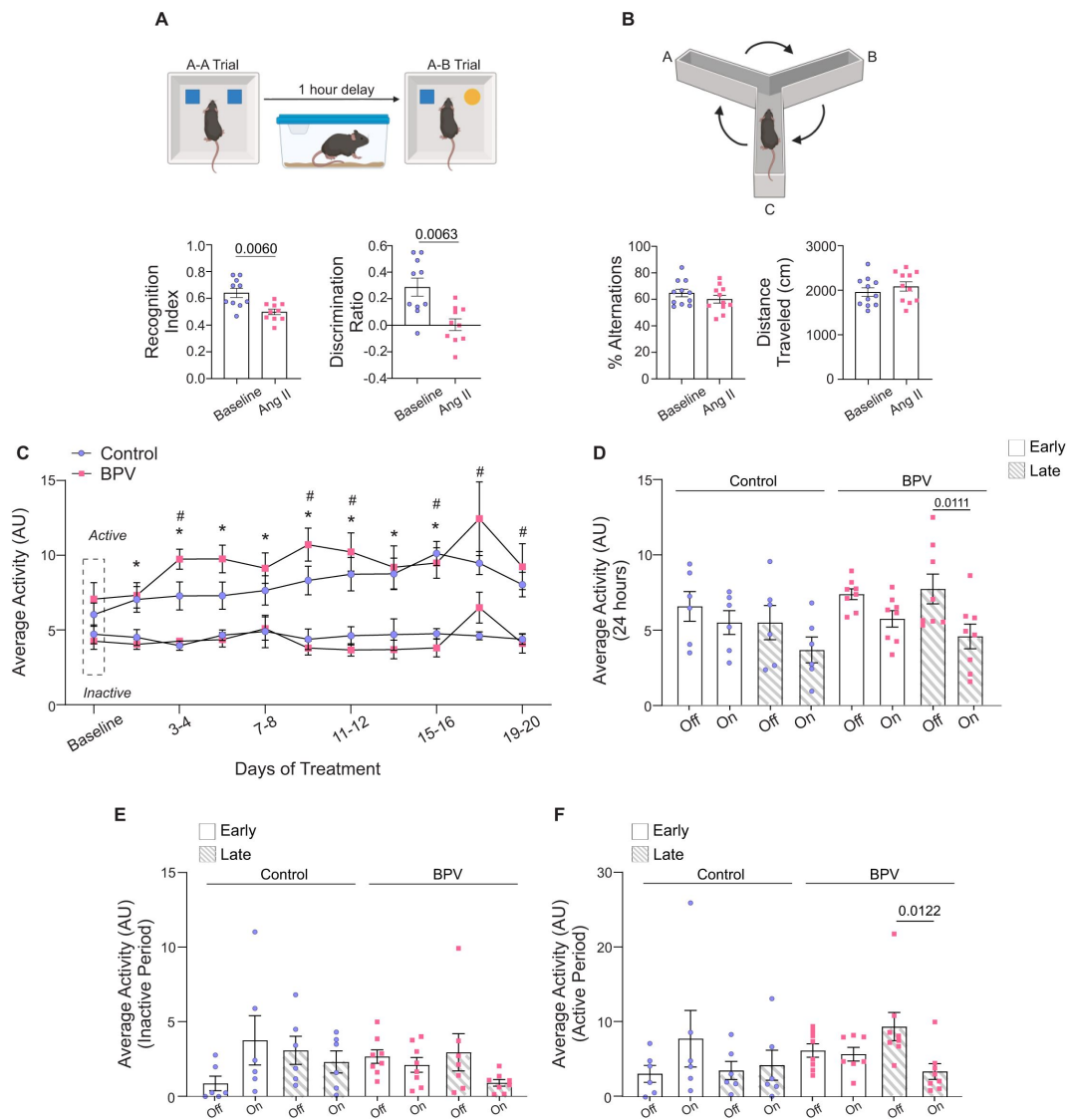


Figure 7.

Behavior and altered cognitive function of BPV mice.

A, Schematic of the novel object recognition test with a 1-hour delay between the A-A and A-B trials (top), accompanied by summarized recognition and discriminatory index data for mice during the saline infusion period (baseline) and following 25 days of pulsatile Ang II infusion (Ang II). **B**, Diagram of 10-minute spontaneous Y-maze experimental setup (top) and the summary of % alternation and distance traveled during baseline and after 25 days of Ang II treatment (bottom). **C**, Summary data of two-day, 24-hour activity averages throughout treatment during active and inactive periods, with the Baseline phase (~5-day saline infusion) marked by the dashed rectangle. **D**, Summary data of 24-hour averaged activity recorded when the infusion pump was *on* vs *off* during Early (days 3-5) and Late (days 23-25) treatment phases. **(E)** Summary of activity when the pump is *on* vs *off* during the Early and Late phases of treatment for the inactive cycle. **(F)** Summary of activity when the pump is *on* vs *off* during the Early and Late phases of treatment for the active cycle. Paired t-test (A, $n = 10$ BPV mice), (B, $n = 11$ BPV mice). Two-way ANOVA repeated measures followed by Tukey's comparison test (C, $n = 13$ control mice, $n = 8$ BPV mice) (D-F, $n = 6$ control mice, $n = 8$ BPV mice). “*” denotes $p < 0.05$ vs Baseline for control, active period. “\$” and “Δ” denote $p < 0.05$ (active vs inactive period) for control and BPV, respectively.

AU=arbitrary units, **BPV**=blood pressure variability

hypertension¹¹, prolonged BPV beyond the 25-day treatment period could potentially lead to hypertension. Furthermore, we anticipate that sustained BPV and its resultant hypertension may drive cardiac remodeling, specifically, left ventricular hypertrophy. These findings underscore the need for future studies to investigate the long-term effects of BPV on cardiac morphology.

We explored norepinephrine as an alternative agent to induce BP variations, **Figure 4**. Similar to Ang II-infused mice, NE-treated mice exhibited a significant increase in BPV (i.e. increased CV) while maintaining minimal changes in the 24-hr averaged MAP. However, NE-induced BP pulses elicited a tachycardic response likely driven by NE stimulation of cardiac β 1-adrenergic receptors⁴³. Notably, chronic NE administration exaggerated this tachycardic response, with a steeper positive slope over time (Early vs Late; $P < 0.0001$) during the Inactive period, **Figure 4D**. Additionally, NE-treated mice exhibited signs of lethargy, impaired mobility, tachypnea, and an increased rate of premature death beginning on day 11 of treatment. These findings emphasize the need for further investigation into alternative BPV pressors, as each may exert distinct off-target effects.

Our model underscores the importance of assessing BPV independently of average BP values. Retrospective studies in primary care patients reveal BPV occurrences in both hypertensive and non-hypertensive adults⁴⁴, establishing BPV as a critical yet modifiable risk factor, often overlooked by standard in-office single BP measurements. In our model, the significantly higher 24-hour average PP in BPV mice suggests that PP is the most sensitive BP-related variable. Compared to controls, BPV mice exhibited persistently lower DBP and higher PP during the inactive period, indicative of a compensatory mechanism that maintained 24-hour MAP within physiological ranges. However, other factors mitigating hypertension onset cannot be ruled out.

The autonomic nervous system (ANS) tightly regulates BP via changes in heart rate, systemic vascular resistance, and stroke volume. Notably, the baroreflex plays a critical role in maintaining cerebral perfusion pressure⁴⁷. Acute BP elevations activate arterial baroreceptors, which inhibit sympathetic outflow (via enhanced vagal activity) and reduce HR and peripheral vascular resistance⁴⁷, thereby lowering arterial BP. However, to sustain relatively stable CBF, baroreflex-mediated inhibition of sympathetic outflow and arteriole dilation is counterbalanced by the intrinsic myogenic properties of the cerebral vasculature, which constricts in response to increases in arterial BP. Dysregulation of these dynamic processes heightens the risk of both cerebral hypoperfusion and hyperperfusion. Notably, impaired baroreflex sensitivity has been linked to aging⁴⁸ and mild cognitive impairments⁴⁹.

In our model, chronic BPV significantly blunted the bradycardic response, suggestive of ANS dysregulation, potentially driven by enhanced sympathetic activity, reduced parasympathetic activity, or a combination of both. Previous studies indicate that chronic Ang II infusion suppresses and resets baroreceptor sensitivity independently of the pressure effect, with partial reversal occurring within 30 minutes post-Ang II infusion⁵⁰. Unlike conventional sustained Ang II infusion approaches (osmotic pump models), our protocol delivers Ang II intermittently at 3- or 4-hour intervals. This regimen likely mitigates desensitization and Ang II receptor internalization, a response well-documented in sustained infusion models^{51,52}. Furthermore, the same Ang II dose evoked comparable BP increases at the start and end of treatment. Thus, our results support diminished baroreflex responses, aligning with previous reports linking elevated BPV to autonomic dysregulation⁵³, though the underlying cellular mechanisms remain poorly understood.

Cerebral autoregulation serves as a protective mechanism against fluctuations in cerebral perfusion pressure²². Vessels upstream from cerebral capillaries, including pial arteries^{54–56} and larger extracranial vessels^{57–59}, help buffer arterial pressure changes, thereby limiting excessive variations in CBF. In response to chronic and sustained hypertension, parenchymal arterioles exhibit enhanced reactivity to pressure increases^{21,60,61}. This is

further accompanied by arteriole remodeling^{21,62,63}. Although our model does not involve sustained hypertension, we present *in vivo* evidence of heightened myogenic responses of parenchymal arterioles to both BP elevations and reductions. These observations are particularly intriguing as they suggest potential adaptive mechanisms in response to large blood pressure fluctuations.

Chronic BP elevations may drive upregulation of voltage-dependent calcium channels (VDCC), or suppression of K⁺ currents⁶⁴. Cipolla et al., reported that parenchymal arterioles are approximately 30-fold more sensitive to VDCC inhibition via nifedipine than middle cerebral arteries⁶⁵. Notably, VDCC expression increases in hypertension^{66–72}, and long-acting calcium channel blockers (i.e., amlodipine), are among the most effective treatments for BPV^{73,74}. In an effort to mitigate excessive vasoconstriction and optimize blood flow, during periods of BP-induced constrictions, vasodilatory signals may accumulate. Consequently, when BP decreases, parenchymal arterioles in BPV mice may exhibit greater dilation compared to controls, **Supplemental Figure 5F**⁷⁵. Augmented vasodilation has been reported as adaptive response in rodent models of chronic hypoperfusion^{75,76}. The dynamic nature of BP fluctuations in our model (i.e. increases and decreases) uniquely allows observations of multiple processes occurring concurrently (e.g. enhanced vasoconstrictions in response to high BP, and enhanced dilations potentially resulting from vasoconstriction-evoked ischemia).

If this interpretation holds, our model may reflect conditions in which parenchymal arterioles experience both high pressure and ischemia. The resulting heightened myogenic responses may amplify diameter changes, potentially diminishing pressure buffering capacity and increasing vulnerability to extreme BP fluctuations, predisposing brain tissue to episodic ischemia and/or hyperperfusion. Future studies addressing the impact of BPV on capillary perfusion may provide further insights into the link between BPV and brain pathology, including microbleeds^{77,78}.

Chronic BPV significantly blunted the NVC response to whisker stimulation. In control mice, NVC exhibited a pressure-dependent effect, with greater response magnitudes observed at higher arterial BP levels—an effect that was abolished in BPV mice. One possible explanation is that the enhanced NVC response (under high BP conditions) in control mice stems from greater baseline arteriole tone, facilitating a more efficient dilatory response to vasoactive signals released during sensory-evoked stimuli. Conversely, the heightened reactivity of parenchymal arterioles to intravascular pressure in BPV mice (**Figure 5H**⁷⁹) suggests a predominant vasoconstrictive state⁷⁹, potentially accompanied by reduced availability of vasodilatory mediators involved in NVC (e.g. nitric oxide, epoxyeicosatrienoic acids (EETs), cyclooxygenase-derived prostaglandins)^{80,81}. Thus, the blunted NVC response observed during elevated BP periods in BPV mice may also result from impaired vasodilation.

Additionally, upon termination of WS, vasoconstrictive mechanisms—exacerbated by elevated BP are unmasked, leading to a faster recovery in arteriole diameter post-stimulus⁸². Another potential explanation is a neuronal mechanism, where diminished neural activity or reduced release of vasodilatory signals may contribute to the impaired NVC response. Further studies should investigate the impact of high BPV on the neurovascular complex under awake conditions, recognizing that factors such as activity and arousal^{83,84}, can elevate BP and potentially confound pressure-dependent neurovascular effects.

Beyond blunting NVC, our study demonstrates that chronic BPV contributes to cognitive decline, as evidenced by decreased discrimination and recognition indices in the NOR test. Clinical studies have similarly linked elevated BPV to impairments across multiple cognitive domains, including memory and executive function^{85,86}. Post hoc analyses of the SPRINT MIND trial revealed that patients with elevated BPV experienced the most rapid decline in processing speed⁸⁶. Additionally, findings from the Alzheimer's Disease Neuroimaging Initiative indicated that greater SBP variability correlated with poorer episodic memory performance⁸⁵.

To assess cognitive function, we employed the NOR test to evaluate episodic memory⁸⁷ and the spontaneous Y maze test to assess short-term spatial memory⁸⁸. Because telemetry systems necessitated individual housing to avoid signal interference, all mice used in this study were single-housed. Previous studies on C57BL/6 mice have reported conflicting effects of single housing on learning and memory^{89–91}, with some suggesting stress-induced deficits and others reporting no impact^{92–95}. These discrepancies may stem from variables such as age at isolation onset, duration of isolation, and behavioral test conditions. To minimize confounding factors, we implemented stringent efforts to ensure consistency across all groups, including standardized cage changes, food replenishment, and pump refills conducted on a single day between 1:00 and 4:00 PM. Our results indicate that while 25 days of chronic BPV does not affect short-term spatial memory, it significantly impairs episodic memory.

Conclusions

Our study demonstrates that high BPV disrupts cellular communication in the neurovascular complex, contributing to cognitive decline. Importantly, our model elicited substantial BP fluctuations while preventing the onset of hypertension, providing direct evidence of BPV's causal role in impaired neurovascular function. These findings support the inclusion of BPV assessment as a critical screening and diagnostic tool alongside conventional BP measurements to address cardiovascular and neurovascular risks. Moreover, the adaptable murine model developed in this study offers a robust framework for future investigations into BPV's impact on brain function.

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Additional files

Supplemental Figure 1. Exclusion criteria for myogenic response analysis. **A-B**, Representative raw traces (from one mouse) of MAP (top) and parenchymal arteriole diameter (bottom) showing WS-evoked dilations (**A**) and random dilatory events (**B**). Grey dashed rectangles outline the period when the pump is on and BP transitions to higher values (Ang II infusion). **C-D**, Expanded data corresponding to the dashed rectangle shown in (**A-B**), with dilatory responses to WS outlined within green dashed rectangles (**C**) and transient dilatory events shown within orange dashed rectangles (**D**). **E-F**, Representative raw traces of MAP (top) and parenchymal arteriole diameter (bottom) following removal of sensory-evoked dilations or the random dilatory events outlined in (**D**). **G-H**, Scatter plot and resultant linear regression line of the filtered MAP vs parenchymal diameter corresponding to (**E-F**). MAP = mean arterial pressure

Supplemental Figure 2. Seasonal effects on mean arterial pressure. **A**, Summarized data of 24-hr MAP at baseline (5 days saline infusion) in mice with (+) or without (-) a cranial window. **B**, Summarized data of two-day average MAP at baseline and 20 days of treatment. Spring-Summer corresponds to warmer months (May-September), and Fall-Winter corresponds to colder months (October-April). Mann-Whitney test (**A**, $n = 14$ mice with craniotomy, 6 without craniotomy). Two-way ANOVA repeated measures followed by Sidak's multiple comparisons test (**B**, $n = 11$ mice in Spring-Summer, 9 mice in Fall-Winter). MAP = mean arterial pressure

Supplemental Figure 3. One-hour averages of MAP over 24hrs-Circadian Profile. **A**, Representative raw trace of one-hour average MAP for 24hrs of all control mice and BPV mice (**B**), with controls summarized in (**C**) and BPV in (**D**). Inactive corresponds to daylight hours (6am-6pm). (A-D, $n = 13$ control mice, $n = 8$ BPV mice). **BPV**=blood pressure variability, **MAP** = mean arterial pressure [↗](#)

Supplemental Figure 4. Transient Ang II infusions induced pulsatile blood pressure in control mice. Summary data of averaged BP (5 minutes) while the pump is Off or On and infusing Ang II in controls. Data was extracted from the inactive (daytime) period at days 3-5 of treatment protocol. Paired Wilcoxon test ($n = 6$ control mice). **DBP** = diastolic blood pressure, **MAP** = mean arterial pressure, **PP** = pulse pressure, **SBP** = systolic blood pressure. [↗](#)

Supplemental Figure 5. Directional myogenic responses. **A**, Summary data for averaged BP corresponding to the Low-to-High BP transition. **B**, Representative raw trace showing MAP (top) and parenchymal arteriole diameter (bottom) during Low-to-High MAP transition shaded in orange (left) and the corresponding representative scatter plot of MAP vs parenchymal arteriole diameter (right). **C**, Summary of MAP-diameter linear regression slopes corresponding to Low-to-High MAP transition. **D**, Summary data of the average high and low MAP recorded immediately following cessation of pump infusion and blood pressure transitions from high to low BP (High-to-Low). **E**, Representative raw trace showing MAP (top) and parenchymal arteriole diameter (bottom) during High-to-Low MAP transition shaded in green (left) and the corresponding representative scatter plot of MAP vs parenchymal arteriole diameter (right). **F**, Summary of MAP-diameter linear regression slopes corresponding to High-to-Low MAP transition. Two-way ANOVA repeated measures followed by Sidak's multiple comparisons test (A, $n = 12$ runs/7 control mice, $n = 14$ runs/8 BPV mice) (D, $n = 10$ runs/8 control mice, $n = 12$ runs/7 BPV mice). Unpaired t-test and Mann-Whitney test (C, $n = 12$ runs/7 control mice, $n = 14$ runs/8 BPV mice) (F, $n = 10$ runs/8 control mice, $n = 12$ runs/7 BPV mice). **BPV**=blood pressure variability, **MAP** = mean arterial pressure [↗](#)

Supplemental Table 1. Pulsatile Ang II infusions induced an increase in blood pressure variability (BPV) measured by coefficient of variance. **A-B**, Summary data of calculated two-day average coefficient of variance (CV) of MAP, SBP, DBP, and PP during the active period (**A**) and inactive period (**B**). CV was calculated as the ratio of the standard deviation of hourly BP averages to 24-hour mean blood pressure. Two-way ANOVA repeated measures followed by Dunnett's comparison test; $n = 13$ control mice, $n = 8$ BPV mice. “*” denotes $p < 0.05$ vs Baseline and “#” denotes $p < 0.05$ between groups. **BPV** = blood pressure variability, **CV**, coefficient of variance, **DBP** = diastolic blood pressure, **MAP** = mean arterial pressure, **PP** = pulse pressure, **SBP** = systolic blood pressure. [↗](#)

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Reviewer #1 (Public review):

This study examined the effect of blood pressure variability on brain microvascular function and cognitive performance. By implementing a model of blood pressure variability using intermittent infusion of AngII for 25 days, the authors examined different cardiovascular variables, cerebral blood flow and cognitive function during midlife (12-15-month-old mice). Key findings from this study demonstrate that blood pressure variability impairs baroreceptor reflex and impairs myogenic tone in brain arterioles, particularly at higher blood pressure. They also provide evidence that blood pressure variability blunts functional hyperemia and impairs cognitive function and activity. Simultaneous monitoring of cardiovascular parameters, in vivo imaging recordings, and the combination of physiological and behavioral studies reflect rigor in addressing the hypothesis. The experiments are well designed, and data generated are clear.

A number of issues raised earlier were addressed by the authors in the revised manuscript. The responses are convincing. These included circadian rhythm considerations, baroreflex findings, BP fluctuations driven by animal movement, and data presentation.

Overall, this is a solid study with huge physiological implications. I believe that it will be of great benefit to the field.

<https://doi.org/10.7554/eLife.104082.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

This study examined the effect of blood pressure variability on brain microvascular function and cognitive performance. By implementing a model of blood pressure variability using an intermittent infusion of AngII for 25 days, the authors examined different cardiovascular variables, cerebral blood flow, and cognitive function during midlife (12-15-month-old mice). Key findings from this study demonstrate that blood pressure variability impairs baroreceptor reflex and impairs myogenic tone in brain arterioles, particularly at higher blood pressure. They also provide evidence that blood pressure variability blunts functional hyperemia and impairs cognitive function and activity. Simultaneous monitoring of cardiovascular parameters, in vivo imaging recordings, and the combination of physiological and behavioral studies reflect rigor in addressing the hypothesis. The experiments are well-designed, and the data generated are clear. I list below a number of suggestions to enhance this important work:

(1) Figure 1B: It is surprising that the BP circadian rhythm is not distinguishable in either group. Figure 2, however, shows differences in circadian rhythm at different timepoints during infusion. Could the authors explain the lack of circadian effect in the 24-h traces?

The circadian rhythm pattern is apparent in Figure 2 (Active BP higher than Inactive BP), where BP is presented as 12-hour averages. When the BP data is expressed as one-hour averages (rather than minute-to-minute) over 24 hours, now included in the revised manuscript as Supplemental Figure 3C-D, the circadian rhythm becomes noticeable. In addition, we have included one-hour average BP data for all mice in the control and BPV groups, Supplemental Figure 3A-B.

Notably, the Ang-II induced pulsatile BP pattern remains evident in the one-hour averages for the BPV group, Supplemental Figure 3B. To minimize bias and validate variability, pump administrations start times were randomized for both control and BPV groups, Supplemental Figure 3A-B. Despite these adjustments, the circadian rhythm profile of BP is consistently maintained across individual mice and in the collective dataset, Supplemental Figure 3C-D.

(2) While saline infusion does not result in elevation of BP when compared to Ang II, there is an evident "and huge" BP variability in the saline group, at least 40mmHg within 1 hour. This is a significant physiological effect to take into consideration, and therefore it warrants discussion.

Thank you for this comment. The large variations in BP in the raw traces during saline infusion reflects transient BP changes induced by movement/activity, which is now included in Figure 1B (maroon trace). The revised manuscript now includes Line 222 "Note that dynamic activity-driven BP changes were apparent during both saline- and Ang II infusions, Figure 1B".

(3) The decrease in DBP in the BPV group is very interesting. It is known that chronic Ang II increases cardiac hypertrophy, are there any changes to heart morphology, mass, and/or function during BPV? Can the decrease in DBP in BPV be attributed to preload dysfunction? This observation should be discussed.

The lower DBP in the BPV group was already present at baseline, while both groups were still infused with saline, and was a difference beyond our control. However, this is an important and valid consideration, particularly considering the minimal yet significant increase in SBP within the BPV group (Figure 1D). Our goal was to induce significant transient blood pressure responses (BPV) and investigate the impact on cardiovascular and neurovascular outcomes in the absence of hypertension. We did not anticipate any major cardiac remodeling at this early time point (considering the absence of overt hypertension) and thus cardiac remodeling was not assessed and this is now discussed in the revised manuscript (Line 443-453).

(4) Examining the baroreceptor reflex during the early and late phases of BPV is quite compelling. Figures 3D and 3E clearly delineate the differences between the two phases. For clarity, I would recommend plotting the data as is shown in panels D and E, rather than showing the mathematical ratio. Alternatively, plotting the correlation of ΔHR to ΔSBP and analyzing the slopes might be more digestible to the reader. The impairment in baroreceptor reflex in the BPV during high BP is clear, is there any indication whether this response might be due to loss of sympathetic or gain of parasympathetic response based on the model used?

We appreciate the reviewer's suggestion and have accordingly generated new figures displaying scatter plots of SBP vs HR with linear regression analysis (Figure 3D-G). Our goal is to further investigate which branch of the autonomic nervous system is affected in this model. The loss of a bradycardic response suggests either an enhancement of sympathetic activity, a reduction in parasympathetic activity, or a combination of both. This is briefly discussed in the revised manuscript (Line 486-496).

Heart rate variability (HRV) serves as an index of neurocardiac function and dynamic, non-linear autonomic nervous system processes, as described in Shaffer and Ginsberg[1]. However, given that our data was limited to BP and HR readings collected at one-minute intervals, our primary assessment of autonomic function is limited to the bradycardic response. Further studies will be necessary to fully characterize the autonomic parameters influenced by chronic BPV.

(5) Figure 3B shows a drop in HR when the pump is ON irrespective of treatment (i.e., independent of BP changes). What is the underlying mechanism?

We apologize for any lack of clarity. These observed heart rate (HR) changes occurred during Ang II infusion, when blood pressure (BP) was actively increasing. In the control group, the pump solution was switched to Ang II during specific periods (days 3-5 and 21-25 of the treatment protocol) to induce BP elevations and a baroreceptor response, allowing direct comparisons between the control and BPV group.

To clarify this point, we have revised Line 260-263 of the manuscript: “To compare pressure-induced bradycardic responses between BPV and control mice at both early and later treatment stages, a cohort of control mice received Ang II infusion on days 3-5 (early phase) (Supplemental Figure 4) and days 21-25 (late phase) thereby transiently increasing BP”.

Additionally, a detailed description has been added to the Methods section (Line 96-101): “Controls receiving Ang II: To facilitate between-group comparisons (control vs BPV), a separate cohort of control mice were subjected to the same pump infusion parameters as BPV mice but for a brief period receiving Ang II infusions on days 3-5 and 21-25 for experiments assessing pressure-evoked responses, including bradycardic reflex, myogenic response, and functional hyperemia at high BP.”

(6) The correlation of Δ diameter vs MAP during low and high BP is compelling, and the shift in the cerebral autoregulation curve is also a good observation. I would strongly recommend that the authors include a schematic showing the working hypothesis that depicts the shift of the curve during BPV.

Thank you for this insightful comment. The increase in vessel reactivity to BP elevations in parenchymal arterioles of BPV mice suggests that chronic BPV induces a leftward shift and a potential narrowing of the cerebral autoregulation range (lower BP thresholds for both the upper and lower limits of autoregulation). This has been incorporated (and discussed) into the revised manuscript (see Figure 5N).

One potential explanation for these changes is that the absence of sustained hypertension, a prominent feature in most rodent models of hypertension, limits adaptive processes that protect the cerebral microcirculation from large BP fluctuations (e.g., vascular remodeling). While this study does not specifically address arteriole remodeling, the lack of such adaptation may reduce pressure buffering by upstream arterioles, thereby rendering the microcirculation more vulnerable to significant BP fluctuations.

The unique model allows for measurements of parenchymal arteriole reactivity to acute dynamic changes in BP (both an increase and decrease in MAP). Our findings indicate that chronic BPV enhances the reactivity of parenchymal arterioles to BP changes—both during an increase in BP and upon its return to baseline, Supplemental Figure 5C, F. The data suggest an increased myogenic response to pressure elevation, indicative of heightened contractility, a common adaptive process observed in rodent models of hypertension[2-4]. However, our model also reveals a notable tendency for greater dilation when the BP drops, Supplemental Figure 5F. This intriguing observation may suggest ischemia during the vasoconstriction phase (at higher BP), leading to enhanced release of dilatory signals, which subsequently manifest as a greater dilation upon BP reduction. This phenomenon bears similarities to chronic hypoperfusion models[5,6], where vasodilatory mechanisms become more pronounced in response to sustained ischemic conditions. Future studies investigating the effects of BPV on myogenic responses and brain perfusion will be a priority for our ongoing research.

(7) Functional hyperemia impairment in the BPV group is clear and well-described. Pairing this response with the kinetics of the recovery phase is an interesting observation. I suggest elaborating on why BPV group exerts lower responses and how this links to the rapid decline during recovery.

Based on the heightened reactivity of BPV parenchymal arterioles to intravascular pressure (Figure 5), we anticipate that the reduction of sensory-evoked dilations results from an increased vasoconstrictive activity and/or a decreased availability of vasodilatory signaling pathways (NO, EETs, COX-derived prostaglandins)[7,8]. Consequently, the magnitude of the FH response is blunted during periods of elevated BP in BPV mice.

Additionally, upon termination of the stimulus-induced response—when vasodilatory signals would typically dominate—vasoconstrictive mechanisms are rapidly engaged (or unmasked), leading to quicker return to baseline. This shift in the balance between vasodilatory and vasoconstrictive forces favors vasoconstriction, contributing to the altered recovery kinetics observed in BPV mice. This has been included in the Discussion section of the revised manuscript.

(8) The experimental design for the cognitive/behavioral assessment is clear and it is a reasonable experiment based on previous results. However, the discussion associated with these results falls short. I recommend that the authors describe the rationale to assess recognition memory, short-term spatial memory, and mice activity, and explain why these outcomes are relevant in the BPV context. Are there other studies that support these findings? The authors discussed that no changes in alternation might be due to the age of the mice, which could already exhibit cognitive deficits. In this line of thought, what is the primary contributor to behavioral impairment? I think that this sentence weakens the conclusion on BPV impairing cognitive function and might even imply that age per se might be the factor that modulates the various physiological outcomes observed here. I recommend clarifying this section in the discussion.

We thank the reviewer for this comment. Clinical studies have demonstrated that patients with elevated BPV exhibit impairments across multiple cognitive domains, including declines in processing speed[9] and episodic memory[10]. To evaluate memory function, we utilized behavioral tests: the novel object recognition (NOR) task to assess episodic memory[11] and the spontaneous Y-maze to evaluate short-term spatial memory[12].

Previous research indicates that older C57Bl6 mice (14-month-old) exhibit cognitive deficits compared to younger counterparts (4- and 9-month-old)[13]. To ensure rigorous selection for behavioral testing, we conducted preliminary NOR assessment, evaluating recognition memory at the one-hour delay but observing failures at the four-, and 24-hour delays, indicating age-related deficits. Based on these results, animals failing recognition criteria were excluded from subsequent behavioral assessment. However, because no baseline cognitive testing was conducted for the spontaneous Y-maze, it is possible that some mice with aged-related deficits were included in this test, which may have influenced data interpretation.

Additionally, the absence of differences in the Y-maze performance may suggest that short-term spatial memory remains intact following 25 days of BPV, a point that is now discussed in the revised manuscript.

(9) Why were only male mice used?

We appreciate this comment and acknowledge the importance of conducting experiments in both male and female mice. Studies involving female mice are currently ongoing, with

telemetry data collection approximately halfway completed and two-photon imaging studies on functional hyperemia also partially completed. However, using middle-aged mice for these experiments has proven challenging due to high mortality rates following telemetry surgeries. As a result, we initially limited our first cohort to male mice.

(10) In the results for Figure 3: "Ang II evoked significant increases in SBP in both control and BPV groups;...". Also, in the figure legend: "B. Five-minute average HR when the pump is OFF or ON (infusing Ang II) for control and BPV groups...." The authors should clarify this as the methods do not state a control group that receives Ang II.

Please refer to response to comment 5.

Reviewer #2 (Public review):

Summary:

Blood pressure variability has been identified as an important risk factor for dementia. However, there are no established animal models to study the molecular mechanisms of increased blood pressure variability. In this manuscript, the authors present a novel mouse model of elevated BPV produced by pulsatile infusions of high-dose angiotensin II (3.1ug/hour) in middle-aged male mice. Using elegant methodology, including direct blood pressure measurement by telemetry, programmable infusion pumps, in vivo two-photon microscopy, and neurobehavioral tests, the authors show that this BPV model resulted in a blunted bradycardic response and cognitive deficits, enhanced myogenic response in parenchymal arterioles, and a loss of the pressure-evoked increase in functional hyperemia to whisker stimulation.

Strengths:

As the presentation of the first model of increased blood pressure variability, this manuscript establishes a method for assessing molecular mechanisms. The state-of-the-art methodology and robust data analysis provide convincing evidence that increased blood pressure variability impacts brain health.

Weaknesses:

One major drawback is that there is no comparison with another pressor agent (such as phenylephrine); therefore, it is not possible to conclude whether the observed effects are a result of increased blood pressure variability or caused by direct actions of Ang II.

We acknowledge this limitation and have attempted to address the concern by introducing an alternative vasopressor, norepinephrine (NE), Figure 4. A subcutaneous dose of 45 µg/kg/min was titrated to match Ang II-induced transient BP pulse (Systolic BP ~150-180 mmHg), Figure 4A. Similar to Ang II treated mice, NE-treated mice exhibited no significant changes in average mean arterial pressure (MAP) throughout the 20-day treatment period (Figure 4B). Although there was a trend ($P=0.08$) towards increased average real variability (ARV) (Figure 4C left), it did not reach statistical significance. The coefficient of variation (CV) (Figure 4C right) was significantly increased by day 3-4 of treatment ($P=0.02$).

Notably, unlike the bradycardic response observed during Ang II-induced BP elevations, NE infusions elicited a tachycardic response (Figure 4A), likely due to β -1 adrenergic receptor activation. However, significant mortality was observed within the NE cohort: three of six mice died prematurely during the second week of treatment, and two additional mice required euthanasia on days 18 and 20 due to lethargy, impaired mobility, and tachypnea.

While we recognize the importance of comparing results across vasopressors, further investigation using additional vasopressors would require a dedicated study, as each agent

may induce distinct off-target effects, potentially generating unique animal models. Alternatively, a mechanical approach—such as implanting a tethered intra-aortic balloon[14] connected to a syringe pump—could be explored to modulate blood pressure variability without pharmacological intervention. However, such an approach falls beyond the scope of the present study.

Ang II is known to have direct actions on cerebrovascular reactivity, neuronal function, and learning and memory. Given that Ang II is increased in only 15% of human hypertensive patients (and an even lower percentage of non-hypertensive), the clinical relevance is diminished. Nonetheless, this is an important study establishing the first mouse model of increased BPV.

We agree that high Ang II levels are not a predominant cause of hypertension in humans, which is why it is critical that our pulsatile Ang II dosing did not cause overt hypertension, (no increase in 24-hour MAP). Ang II was solely a tool to produce controlled, transient increases in BP to yield a significant increase in BPV.

Regarding BPV specifically, prior studies indicate that primary hypertensive patients with elevated urinary angiotensinogen-to-creatinine ratio exhibit significantly higher mean 24-hour systolic ARV compared to those with lower ratios[15]. However, the fundamental mechanisms driving these harmful increases in BPV remain poorly defined. A central theme across clinical BPV studies is impaired arterial stiffness, which has been proposed to contribute to BPV through reduced arterial compliance and diminished baroreflex sensitivity. Moreover, increased BPV can exert mechanical stress on arterial walls, leading to arterial remodeling and stiffness—ultimately perpetuating a detrimental feed-forward cycle[16].

In our model, male BPV mice exhibited a minimal yet significant elevation in SBP without corresponding increases in DBP, potentially reflecting isolated systolic hypertension, which is strongly associated with arterial stiffness[17,18]. Our initial goal was to establish controlled rapid fluctuations in BP, and Ang II was selected as the pressor due to its potent vasoconstrictive properties and short half-life[19].

We appreciate the reviewer's insightful comment and acknowledge the necessity of exploring alternative mechanisms underlying BPV, and independent of Ang II. It is our long-term goal to investigate these factors in further studies.

Recommendations for the authors:

Reviewer #2 (Recommendations for the authors):

(1) How was the dose of Ang II determined? It seems that this dose (3.1ug/hr) is quite high.

The Ang II dose was titrated in a preliminary study to one that induced a significant and transient BP response without increasing 24-hour blood pressure (i.e. no hypertension).

Ang II was delivered subcutaneously at 3.1 µg/hr, a concentration comparable to high-dose Ang II administration via mini-osmotic pumps (~1700 ng/kg/min)[20], with one-hour pulses occurring every 3-4 hours. With 6 pulses per day, the total daily dose equates to 18.6 µg/day in a ~30 gram mouse.

For comparison, if the same 18.6 µg/day dose were administered continuously via a mini-osmotic pump (18.6 µg/0.03kg/1440min), the resulting dosage would be approximately 431 ng/kg/min[21,22], aligning with subpressor dose levels. Thus, while the total dose may appear high, it is not delivered in a constant manner but rather intermittently, allowing for controlled, rapid variations in blood pressure.

(2) Were behavioral studies performed on the same mice that were individually housed? Individual housing causes significant stress in mice that can affect learning and memory tasks (PMC6709207). It's not a huge issue since the control mice would have been housed the same way, but it is something that could be mentioned in the discussion section.

Behavioral studies were performed on mice that were individually housed following the telemetry surgery. The study was started once BP levels stabilized, as mice required several days to achieve hemodynamic stability post-surgery. Consequently, all mice were individually housed for several days before undergoing behavioral assessment.

To account for potential cognitive variability, earlier novel object recognition (NOR) tests were conducted to established cognitive capacity, and mice that did not meet criteria were excluded from further behavioral testing. However, we acknowledge that individual housing induces stress, which can influence learning and memory, and this is a factor we were unable to fully control. Given that both experimental and control groups experienced the same housing conditions, this stress effect should be comparable across cohorts. A discussion on this limitation is now included in the text.

(3) It looks like one control mouse that was included in both Figures 1 and 2 (control n=12) but was excluded in Table 1 (control n=11), this isn't mentioned in the text - please include the exclusion criteria in the manuscript.

We apologize for the typo—12 control animals were consistently utilized across Figure 1-2, Table 1, Supplemental Table 1, Figure 6C, and Supplemental Figure 2B. Since the initial submission, one control mouse was completed and included into the telemetry control cohort. Thus, in the updated manuscript, we have corrected the control sample size to 13 mice across these figures ensuring consistency.

Additionally, exclusion criteria have now been explicitly included in the manuscript (Line 173-175). Mice were excluded from the study if they died prematurely (died prior to treatment onset) or mice exhibited abnormally elevated pressure while receiving saline, likely due to complications from telemetry surgery.

(4) Please include a statement on why female mice were not included in this study.

As discussed in our response to Reviewer #1, our initial intention was to include both male and female mice in this study. However, high mortality rates following telemetry surgeries significantly constrained our ability to advance all aspects of the study. As a result, we limited our first cohort to males to establish the basics of the model. A statement is now included in the manuscript, Line 50-53: “Female mice were not included in the present study due to high post-surgery mortality observed in 12-14-month-old mice following complex procedures. To minimized confounding effects of differential survival and to establish foundational data for this model, we restricted the investigation to male mice.”

Potential sex differences might be complex and warrants a separate future research to comprehensively assess sex as a biological variable, which are currently ongoing.

(5) On page 14, "experiments from control vs experimental mice were not equally conducted in the same season raising the possibility for a seasonal effect" - does this mean that control experiments were not conducted at the same time as the Ang II infusions in BPV mice? This has huge implications on whether the effects observed are induced by treatment or just batch seasonal effects.

We fully acknowledge the reviewer's concern, and our statement aims to provide transparency regarding the study's limitations. Several challenges contributed to this outcome, including high mortality rates following surgeries (primarily telemetry implantation) and technical issues related to instrumentation, particularly telemetry functionality.

Differences between BPV and saline mice emerge primarily due to mortality or telemetry failures—some mice did not survive post-surgery, while others remain healthy but had non-functional telemeters. This issue was particularly pronounced in 14-month-old mice, as their fragile vasculature occasionally prevented proper BP readings.

Each experiment required a minimum of two and a half months per mouse to complete, with a cost (also per mouse) exceeding \$1500 USD (\$300 pump, \$175 mouse, \$900 telemeters, per diem, drugs, reagents etc.). Despite our best effort to ensure comparable seasonal/batch data, these logistical and technical constraints prevented perfect synchronization.

To evaluate whether seasonal differences influenced our results, we incorporated additional telemetry data into the control cohort. Of the seven included control mice, six underwent the same treatment but were allocated to a separate branch of the study, which endpoints did not require a chronic cranial window. We found no significant differences in 24-hour average MAP during the baseline period between control mice with or without a cranial window, Supplemental Figure 2A. Additionally, we grouped mice into seasonal categories based on Georgia's climate: "Spring-Summer" (May-September) and "Fall-Winter" (October-April) but observed no BP differences between these periods, Supplemental Figure 2B.

Given the absence of seasonal effects on BP and the fact that mice were sourced from two independent suppliers (Jackson Laboratory and NIA), we anticipate that the observed results are driven by treatment rather than seasonal or batch effects.

(6) Methods, two-photon imaging: did the authors mean "retro-orbital" instead of "intra-orbital" injection of the Texas red dye? Also, is this a Texas red-dextran? If so, what molecular weight?

Thank you for this comment. The correct terminology is "retro-orbital" rather than "intra-orbital" injection. Additionally, we utilized Texas Red-dextran (70 kDa, 5% [wt/vol] in saline) for the imaging experiments. These details have now been incorporated into the Methods section.

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